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RESEARCH ARTICLE

Multidimensional Analysis of the Diagnostic Performance of Machine Learning Models in Disease Classification: Implications for Clinical Decision Support Systems

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ABSTRACT Clinical decision support systems face a fundamental challenge in the form of incomplete medical data. This study presents a methodology for designing an expert system for classifying disease entities according to ICD-10. The system operates under conditions of incomplete clinical data and diagnostic uncertainty. A total of 1655 clinical cases from internal medicine ($n = 616$) and pediatrics ($n = 1,039$) were analyzed. Nineteen classification algorithms were compared. A two-stage diagnostic model (TOP-1 and TOP-3), reflecting clinical practice, was applied. The main hypothesis assumed that machine learning algorithms can effectively compensate for diagnostic uncertainty through mechanisms of differential diagnosis. In internal medicine, Gradient Boosting achieved the highest performance (84.74% accuracy), whereas in pediatrics the Multi-Layer Perceptron performed best (87.20% accuracy). Pediatrics demonstrated higher overall performance (mean TOP-3: 81.99%) than internal medicine (77.43%). The TOP-3 mechanism ensured a probability of identifying the correct diagnosis exceeding 94% in pediatrics and 91% in internal medicine, which substantially enhances the clinical safety of the system. The identification of different optimal algorithms for each specialty confirms the need to tailor modeling strategies to the clinical context. On external validation on 120 cases from a separate facility, treated as the primary benchmark of real-world performance, TOP-1 accuracy was 73.33% (internal medicine) and 78.33% (pediatrics), with TOP-3 accuracy of 86.67% and 90.00% respectively, and SHAP and LIME confirming clinically relevant predictions. The TOP-3 methodology developed constitutes an important contribution to the design of decision support systems under limited data availability.

INDEX TERMS Clinical decision support system, comparative algorithm analysis, diagnostic performance assessment, machine learning, medical diagnostics.

I. INTRODUCTION

Automatic medical diagnostic systems must account for multiple clinical parameters and their interactions. The complexity of this process arises from the multidimensional

nature of diagnostic data and necessitates the analysis of a large number of variables. In clinical practice, a complete diagnostic dataset is rarely available [1]. This constitutes a significant problem, often overlooked in the training and validation of clinical decision support systems. The issue is particularly pronounced in specialties with a broad diagnostic spectrum, such as internal medicine and pediatrics.

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The present study is motivated by an identified knowledge gap in the application of machine learning to medical diagnostics under conditions of incomplete information content of medical histories. Previous research has focused primarily on specific algorithmic applications in narrow medical domains, such as polyp detection in colonoscopy [2], sepsis prediction in intensive care units [3], and diagnostics of selected cancer types [4]. These studies largely ignored the problem of incomplete information content of clinical data. The ability of expert systems to function effectively in the presence of incomplete clinical data is a pervasive challenge in real-world medical practice [5]. Most earlier studies assumed access to complete diagnostic data, and many systems were characterized by low interpretability. Clinicians' trust in machine learning-based decision support systems is essential for their adoption in clinical practice. This trust can be fostered by ensuring explainability in terms of clinical relevance [6].

In response to these limitations, the present study proposes a clinical decision support system that implements inference mechanisms under diagnostic uncertainty and data incompleteness, explicitly reflecting real clinical conditions. The system was developed using real-world medical data comprising 1,655 clinical cases: 1,039 cases from pediatrics and 616 from internal medicine. The datasets differ in terms of symptom profiles and degree of completeness.

The aim of the study was to develop a physician decision support system which, based on the information content of medical histories and with explicit consideration of incomplete input data, generates diagnostic proposals. The main research hypothesis posits that appropriately parameterized ensemble algorithms can effectively compensate for diagnostic uncertainty stemming from incomplete clinical data through the implementation of differential diagnosis mechanisms. The proposed approach constitutes a methodological contribution to the modeling of diagnostic uncertainty and enables more effective inference under conditions of limited data availability.

The implemented system incorporates 19 machine learning models. The results provide insights into the diagnostic performance of individual machine learning models, particularly highlighting the advantages of ensemble algorithms. The findings open new perspectives for the automation and optimization of diagnostic processes in real clinical settings characterized by incomplete data.

The present work distinguishes itself from existing studies in several respects. The developed system operates directly on incomplete clinical data, without the application of advanced imputation methods. A further distinguishing feature is the TOP-1/TOP-3 mechanism, which formalizes differential diagnosis as a probabilistic ranking integrated into the decision-support workflow rather than a post-hoc evaluation metric, whereas existing systems generate a single prediction. Additionally, 19 algorithms were compared on real-world, incomplete data from two medical specialties simultaneously, combining comparative breadth with a realistic clinical context. The diagnostic classification system presented in this

work constitutes the core module of the PARROT AI multimodal medical assistant [7], designed to support primary care physicians through automated documentation and clinical decision support.

In summary, the contribution of this study includes:

- 1) Development of inference mechanisms for general disease diagnostics under conditions of incomplete symptom and test-result data. The developed mechanisms do not incorporate missing data imputation and operate under conditions typical of real-world clinical practice.
- 2) Explicit consideration of diagnostic uncertainty in predicting patient disease. The TOP-1/TOP-3 mechanism enables the generation of a probabilistic ranking of diagnoses, reflecting the clinical process of differential reasoning.
- 3) Validation of the proposed methods on real-world data containing incomplete information obtained during actual patient visits.
- 4) Validation of the developed methods on an external dataset.
- 5) Evaluation of the effectiveness of the developed approach on two datasets comprising interview data from an internist and a pediatrician. This enabled the identification of domain-specific optimal modeling strategies.

The structure of the article is as follows: Section II presents an overview of works related to the research topic. Section III describes the proposed approach. Section IV reports the results of the experiments. The paper concludes with a summary and directions for future research.

II. RELATED WORK

Modern clinical decision support systems (CDSS) exhibit varying capacity to operate under conditions of incomplete clinical information. Their development relies on machine learning algorithms that utilize diagnostic patterns derived from real clinical data.

In the study by [8], an expert system was presented that tested six algorithms (Random Forest (RF), KNN, Support Vector Classifier (SVC), MultinomialNB, Logistic Regression and Decision Tree) on datasets comprising up to 415,712 synthetic patients with 75 symptoms and 22 diagnoses. Although the system achieved accuracy exceeding 99%, it was trained solely on complete synthetic data, without analyzing the impact of incompleteness. This substantially limits its relevance to clinical practice. Similarly, [9] developed an Artificial Intelligence (AI) system employing deep learning and natural language processing (NLP) for pediatric diagnostics using 101.6 million data points from 1,362,559 visits, achieving a median accuracy of 0.90. However, this system was trained on structured and complete medical data. In [10], a comparison of nine ML algorithms for cervical cancer detection was performed. Ensemble methods based on decision trees (RF, LightGBM, and CatBoost) achieved the

highest performance. RF attained classification accuracy of 98.4%, sensitivity of 99.3%, and specificity of 97.6%. In that study, missing values were imputed prior to model training, and class imbalance was corrected using the Synthetic Minority Oversampling Technique (SMOTE) exclusively in the training sets. This may limit the direct comparability of these results with those obtained on real-world clinical data.

The concept of differential diagnosis in automated systems was introduced by [11], who proposed a two-stage CDSS for treatment recommendation in hepatocellular carcinoma. Accuracy increased from 67.27% to 87.27% after incorporating a second therapeutic option, which is clinically significant. Nonetheless, the system did not address the issue of incomplete input data. An ensemble approach with an interpretable model was also presented in the context of cardiovascular disease risk prediction [12]. A stacking ensemble model combining RF, XGBoost, and logistic regression was employed, achieving a ROC-AUC of 0.918. SHAP (SHapley Additive exPlanations) analysis revealed nonlinear interactions among clinical features, supporting the interpretability of predictions and the clinical credibility of the model. In the context of ICD-10 coding, [13] presented a system combining BioBERT with rule-based methods, achieving an F1-score of 0.769 for ICD-10-CM. Its success was largely attributable to the integration of a pre-trained language model with expert-rule-driven preprocessing. A practical clinical implementation was demonstrated by [6], who described an NLP system deployed in a hospital environment for allergy detection, achieving 92.6% sensitivity and 88.8% precision. The system recognizes the problem of incomplete data through the detection of inconsistencies in medical records. However, it does not offer a systematic methodology for modeling diagnostic uncertainty. In the context of processing noisy data from clinical conversations, the study [14] demonstrated that model ensemble-based methods, used in multi-class and multi-label text classification, achieve robust results on real-world medical transcripts, handling the challenges associated with informal language and incomplete utterances characteristic of clinical interviews.

A systematic review of 83 studies by [15] reported an overall diagnostic accuracy of AI models at 52.1%, with AI performing significantly worse than expert clinicians. These results underscore the importance of rigorous evaluation of expert systems. A comprehensive symptom-based evaluation was presented by [16] using 400 standardized clinical cases. The best-performing model achieved 67.5% accuracy in M1, 87.3% in M3 and 90% in M5, demonstrating comparable or superior performance to experienced physicians (mean M1: 61.2%, M3: 72.5%), particularly in differential diagnosis ranking.

The implementation of mechanisms for handling incomplete data remains a critical aspect of expert-system development. Reference [17] proposed a deep learning method for imputing missing values in medical records, accounting for patterns of deficiencies and temporal relationships, achieving

5.3–15.5% lower error rates. Reference [18] identified that missing values in medical data are never entirely random, introducing a systematic nature of data incompleteness.

A systematic review of 111 studies [19] concluded that deep learning methods outperform traditional imputation techniques, though no single approach offers a universally generalizable solution. Methodological approaches to handling missing data were extensively discussed by [20], who reviewed 110 papers and classified methods for dealing with missing values and ambiguous outcomes. Reference [21] proposed neutrosophic logic combined with machine learning to predict chronic kidney disease (CKD), incorporating three membership degrees (true, uncertain, false). Although this method outperformed traditional Machine Learning (ML) algorithms, it assumes availability of the full diagnostic feature set. Reference [22] applied a combination of supervised and unsupervised learning with the Apriori algorithm to 3,490 records, achieving 86.73% accuracy. However, this approach does not systematically address input data incompleteness.

Interpretability in expert systems is illustrated by [23], who proposed a two-layer RF model for diagnosing Alzheimer's disease using 11 data modalities from 1,048 patients. The model achieves 93.95% accuracy with integrated decision-explanation mechanisms. Practical implementations are exemplified by [24], who developed an expert system for diagnosing focal segmental glomerulosclerosis using 15 ML algorithms. The system reached 93% accuracy for LightGBM and RF, reducing diagnostic errors by 13.4%. Reference [25] demonstrated that XGBoost performs better on datasets with missing values than on those with imputed data, showing the advantage of native handling of incomplete information.

An analysis of existing CDSS reveals the following research gaps, which this study aims to address:

First, existing systems typically assume the completeness of input data or rely on imputation techniques before model training begins [1], [10], [26]. No single imputation method provides a universal solution [19], [20]. This work addresses this gap by operating directly on incomplete clinical data without imputation, treating undocumented symptoms as absent in accordance with clinical reasoning practice.

Second, the highest diagnostic performance is achieved in specialized systems limited to specific disease groups [10], [24]; however, results often do not reflect real-world clinical conditions due to the omission of incomplete data or reliance on synthetic datasets [8]. This work evaluates 19 algorithms across 87 diagnostic categories in two medical specialties, using real-world clinical cases with varying degrees of data completeness.

Third, existing differential diagnosis mechanisms do not systematically quantify diagnostic uncertainty [11], and interpretability remains limited in most implemented systems [12], [23]. This work introduces a probability-based TOP-1/TOP-3 ranking mechanism that formalizes differential diagnosis through explicit quantification of uncertainty, supplemented by SHAP and LIME interpretability analysis.

In contrast to recent studies, which typically address a single disease entity on complete or imputed data, optimize a narrow set of algorithms, or report predictions without quantifying diagnostic uncertainty, the present work integrates operation on incomplete data without imputation, a broad comparative evaluation across 19 algorithms and 87 diagnostic categories in two specialties, and a formalized quantification of diagnostic uncertainty within a single decision-support system.

III. RESEARCH METHODOLOGY

The research methodology comprises the following stages:

- 1) Classification and analysis of clinical data incompleteness.
The aim of this stage is to analyze the impact of different types of clinical data incompleteness on the medical diagnostic process.
- 2) Design of the diagnostic classification system architecture.
At this stage, the architecture of the inference mechanism is developed, explicitly accounting for incomplete clinical data.
- 3) Definition of the clinical data processing model.
This stage focuses on developing a data model used for training, validation and testing of machine learning models.
- 4) Development of a mechanism for modeling diagnostic uncertainty.
This stage addresses the main research problem, namely the development of an inference mechanism that incorporates diagnostic uncertainty and incomplete clinical data.
- 5) Implementation and evaluation of the developed method.
The aim of the proposed research methodology is to develop a system to support doctors' decisions, which generates diagnostic suggestions based on the information contained in interviews, taking into account the incompleteness of the input data.

A. CHARACTERISTICS OF THE PROBLEM OF INCOMPLETE DATA IN MEDICAL DIAGNOSTICS

Analysis of different clinical datasets reveals missing data in multiple dimensions. By classifying these deficiencies, the following types of clinical data incompleteness can be distinguished: systemic, cognitive, procedural, temporal, semantic and disproportionate.

- Systemic incompleteness arises from imperfections in data collection processes in EHR (Electronic Health Record) systems. Its consequences include missing diagnostic parameters, incomplete medical histories and incomplete test reports. Patterns of missing data in EHR applications are not entirely random; rather, they show a strong association with the presence of comorbidities in patients [26].

- Cognitive incompleteness is linked to the failure of medical staff to complete all fields in the documentation. This may result from time pressure, oversight or the inability to obtain specific information from the patient. In practice, it may affect up to 89% of examinations [27].
- Procedural incompleteness refers to missing results of specific diagnostic tests that were not ordered or performed due to resource constraints, lack of clinical indications or the physician's decision. Missing values and ambiguous test results may lead to biased outcomes [20].
- Temporal incompleteness stems from time limitations in the diagnostic process. It occurs when a diagnosis is made before all test results are available or when the patient does not attend follow-up examinations. Most clinical datasets contain multiple data types, relatively few samples, and many missing observations [17].
- Semantic incompleteness occurs when data are entered into the system in a way that prevents unambiguous interpretation. This may be caused by non-standard abbreviations, ambiguous terminology or coding errors. Health data in EHR systems are less well organized and standardized than medical imaging data. EHRs contain highly heterogeneous and sometimes inconsistent information, including demographic data, diagnoses, procedures, laboratory results and unstructured free-text clinical notes [28].
- Disproportionate incompleteness is a specific form of missing data, where machine learning models are trained on datasets in which certain groups are more likely to have incomplete observations. This may lead to unfair predictive performance for these populations. The problem can be considered in the context of missing data mechanisms, particularly MAR (missing at random) and MNAR (missing not at random), where the distribution of complete data may differ substantially from that of the target population [29].

Clinical data incompleteness has a multidimensional impact on the diagnostic process, affecting both everyday clinical practice and the implementation of medical decision support systems. Lack of complete data leads to biased estimates of diagnostic accuracy [20] and necessitates the use of statistical methods, including probabilistic approaches, in differential diagnosis. A key mechanism through which data incompleteness affects diagnosis is the restriction of the diagnostic spectrum. This restriction may increase the risk of overlooking the correct diagnosis in clinical differential reasoning [19]. In modeling incomplete clinical data, it is therefore crucial to maintain a sufficiently broad spectrum of diagnostic hypotheses. This broad spectrum underpins the effective implementation of differential diagnostic mechanisms. Furthermore, incompleteness may introduce systematic bias towards particular diagnoses, especially when missing data concern parameters that are critical for differentiating between disease entities with similar symptomatology. Another important aspect is non-uniform symptom

representation. Incomplete data may create a distorted clinical picture characterized by over-representation of certain symptoms and underdocumentation of others. In the context of machine learning algorithms, this increases the risk that the model will learn inadequate associations between symptoms and diagnoses, which can substantially reduce predictive performance in real-world clinical settings.

B. ARCHITECTURE OF THE DIAGNOSTIC CLASSIFICATION SYSTEM

The diagnostic classification system was designed as a multilayer architecture for clinical data processing, integrating the acquisition of medical information with machine learning mechanisms (Figure 1). The system follows a pipeline processing paradigm, in which each transformation stage is optimized for the characteristics of medical data and diagnostic requirements. Table 1 presents a summary of the main modules of the diagnostic classification system.

TABLE 1. Characteristics of the modules of the diagnostic classification system.

Module	Function	Input data	Output data
Data processing	Convert text descriptions of symptoms into standardized binary feature vectors.	Raw clinical data	Binary feature vectors
Training	It trains 19 ML algorithms in two categories: standard and advanced.	Feature vectors	Trained ML models
Optimization	Automatically tunes the hyperparameters of each model using the Optuna algorithm.	Trained models	Optimized models by specialty
Evaluation	Measures of diagnostic performance, including TOP-K, confusion matrices, and ROC-AUC curves.	Optimized models	Diagnostic accuracy metrics
Packaging	Stores models in a unified format with integrity checks and metadata.	Evaluated models	Versioned models with metadata
Reporting	It automatically generates reports in XLSX, PDF, and JSON formats, complete with visualizations.	Evaluation results	Clinical and statistical reports
Deployment	It makes predictions based on real-world data and aggregates results from multiple models.	New clinical data	TOP-1 and TOP-3 diagnoses with ICD-10 codes
Prediction	Provides diagnostics via REST API, real-time prediction, and batch processing.	Diagnostic queries	Diagnoses with probabilities

The diagnostic workflow can be initiated through one of three user interfaces: a graphical user interface (GUI), a command-line interface (CLI), and a web-based application programming interface (Web API). These interfaces enable flexible deployment across a range of clinical environments, from individual medical workstations to distributed hospital systems. Clinical data entered by healthcare personnel include a comprehensive set of information: subjective symptoms reported by the patient, objective findings from physical examinations, and quantitative and qualitative parameters from laboratory tests.

All processing requests are routed to the process orchestrator module, which coordinates data flow between system components and manages the sequencing of processing operations. The orchestrator ensures consistent data processing regardless of the user interface and synchronizes parallel processing paths in subsequent stages.

In the initial processing stage, clinical data are transformed within a dedicated data processing module composed of three components: a symptom encoder, a dataset converter and a feature engineering module.

The symptom encoder transforms textual descriptions of clinical symptoms into a unified binary representation, enabling their use by machine learning algorithms. The dataset converter normalizes the data to ensure quality and consistency, eliminating differences in value ranges between clinical parameters and standardizing the format of input data.

The processed data are then forwarded to the training module, which implements two categories of machine learning algorithms: standard trainers and advanced trainers. The standard trainer category includes the following algorithms: Decision Tree, Extra Tree, RF (Random Forest), Extra Trees (set of additional random trees), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Multilayer Perceptron, Naive Bayes (NB) along with variants Multinomial NB and Bernoulli NB, Logistic Regression (LR), Ridge Classifier, Stochastic Gradient Descent (SGD) Classifier, Linear Discriminant Analysis (LDA), and Quadratic Discriminant Analysis. The advanced trainers comprise gradient boosting algorithms: Gradient Boosting (GB), Histogram-based GB, AdaBoost (adaptive boosting), and Dynamic Ensemble (dynamic ensemble with meta-learning mechanism).

All machine learning models are processed through an optimization module that uses the Optuna algorithm for automatic hyperparameter tuning. The optimization procedure is carried out individually for each model and for each medical specialty, ensuring maximized predictive performance while maintaining robust generalization. The optimizer applies Bayesian hyperparameter optimization with an early stopping mechanism, allowing efficient exploration of the parameter space under constrained computational resources.

Following completion of model training, comprehensive performance evaluation is conducted in a dedicated evaluation module. The module computes a wide range of diagnostic metrics, including standard metrics (accuracy, precision, recall, F1-score), Top-K metrics (accuracy for the k

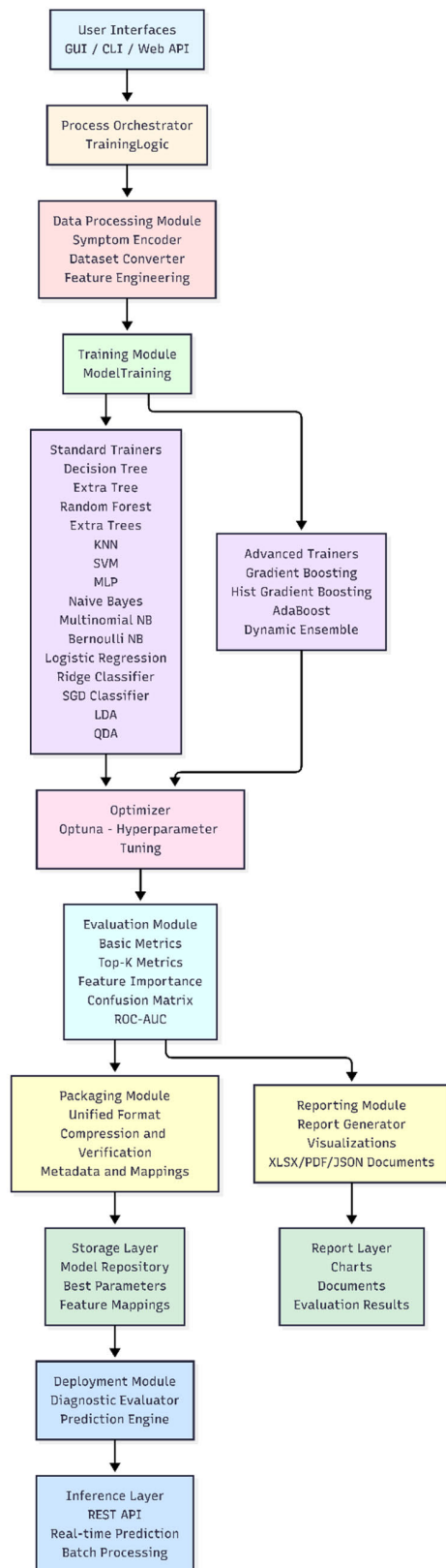


FIGURE 1. Data-flow architecture of the diagnostic classification system.

most probable diagnoses), feature importance analysis, confusion matrices and ROC-AUC curves. Top-K metrics are

particularly relevant in medical applications, as identifying several of the most probable differential diagnoses is critical to the clinical decision-making process. Feature importance analysis supports interpretability of model decisions and enables verification of alignment between learned patterns and established medical knowledge.

Evaluation results are subsequently forwarded in parallel to the packaging and reporting modules. The packaging module implements a unified model storage format, ensuring consistent representation across all machine learning algorithms. Data compression and integrity verification mechanisms are employed to support efficient storage and protection against data corruption. The module also generates metadata describing trained models, including hyperparameter configurations, evaluation outcomes and feature mappings. Trained models and their metadata are stored in a dedicated storage layer that supports model versioning and management of optimal parameter sets for specific medical specialties. This layer also implements feature-mapping mechanisms to ensure consistency between data representations.

Operating concurrently, the reporting module automatically compiles results into formats suitable for both clinical interpretation and further statistical analysis. The report generator produces three categories of outputs: XLSX (Excel Open XML Spreadsheet file format) spreadsheets containing detailed tabular summaries, PDF (Portable Document Format) documents formatted for archival purposes, and JSON (JavaScript Object Notation) files enabling programmatic integration with external systems. In addition, the module generates visualizations such as comparative performance charts, confusion matrices, ROC curves and feature importance plots. All generated documents and visualizations are stored in the reporting layer, enabling centralized management of evaluation results and controlled access for different user groups. After completion of training and evaluation, the trained models are deployed through a deployment module integrating a diagnostic evaluator and a prediction engine. The diagnostic evaluator performs predictions on real-world data using the trained models, while the prediction engine manages inference workflows and aggregation of outputs from multiple models.

The system provides three prediction modes within the prediction layer: a REST API for diagnostic queries from external systems, a real-time prediction mode ensuring low latency for interactive clinical use, and batch processing for efficient analysis of large collections of clinical cases. The system generates a TOP-3 ranking of the most probable diagnoses, mapping predicted class indices to corresponding ICD-10 codes with full disease descriptions. For each suggested diagnosis, a prediction confidence score is calculated, enabling clinicians to assess the credibility of individual diagnostic recommendations.

All classification results are presented through a clear visual interface that facilitates interpretation and effective use in clinical decision-making. The interface also provides diagnostic context for each suggested diagnosis, including

key symptoms supporting the diagnostic hypothesis and indications for potential supplementary examinations.

C. STRUCTURE OF CLINICAL DATA AND PROCESSING WORKFLOW

The data analyzed in this study were derived from electronic medical records of patients treated in two selected medical specialties: pediatrics and internal medicine. These datasets reflect distinct diagnostic profiles and population characteristics. The initial dataset comprised medical documentation from 2,057 clinical cases, including 1,160 cases from pediatric departments and 897 cases from internal medicine units. This composition ensured a representative sample for both specialties analyzed. The documentation contained detailed information on subjective symptoms reported by patients or their caregivers, objective findings identified during physical examination, a limited set of laboratory test results, and final diagnoses coded according to the International Classification of Diseases, 10th Revision (ICD-10). Given the inherent problem of data incompleteness in real-world medical records and the requirement to ensure data quality sufficient for reliable algorithmic analysis, a multi-stage case selection procedure was applied, based on statistical criteria of representativeness commonly used in medical research. The case selection methodology was developed with reference to systematic studies on optimal sample size in machine learning models applied to medicine. Models trained on insufficiently sized groups may be unstable and generate incorrect diagnoses, posing a significant risk to patient safety. Moreover, training sample size may account for up to 49% of the variance in reported algorithm accuracy [30].

The selection process also addressed challenges related to diagnostic class imbalance typical of real-world medical datasets, where the prevalence of disease entities may differ substantially across specialties.

Five selection criteria were applied in the data selection process:

- 1) Presence of a medical diagnosis coded according to ICD-10.
- 2) Exclusion of ICD-10 diagnostic categories irrelevant to real clinical diagnostic practice.
- 3) Minimum availability of diagnostic features per case: at least three symptoms or clinical signs.
- 4) Clinical relevance of medical referrals: exclusion of cases with at least three symptoms in which specialist referrals accounted for at least 50% of all diagnostic parameters.
- 5) Minimum representation of individual diagnostic codes: a threshold of at least three cases per diagnostic code.

After applying all selection criteria, the final dataset consisted of 1,655 clinical cases, including 1,039 cases from pediatrics and 616 from internal medicine. The final dataset retained representativeness for the principal diagnostic categories in both specialties, providing a robust basis for

comparative analyses of machine learning algorithm effectiveness in medical diagnostics.

TABLE 2. Distribution characteristics of diagnostic symptoms in the test dataset.

Statistical parameter	Pediatrics (n=1039)	Internal medicine (n=616)	Difference (P-I) ^a
ALL SYMPTOMS TAKEN INTO ACCOUNT			
Mean ± standard deviation	11.12 ± 5.59	10.24 ± 5.10	+0.88
Median	10.0	9.0	+1.0
Range (min-max)	3–35	3–36	-
POSITIVE SYMPTOMS (TRUE)			
Mean ± standard deviation	9.11 ± 4.46	8.96 ± 4.64	+0.15
Median	9.0	8.0	+1.0
Range (min-max)	0–31	1–29	-
MEASURES OF VARIABILITY AND PROPORTIONS			
Coefficient of variation (%)	48.9%	51.8%	-2.9 p.p.
Proportion of positive symptoms relative to all symptoms (%)	81.9%	87.6%	-5.7 p.p.

^aDifference between pediatrics and internal medicine; p.p.: percentage points; Coefficient of variation = (standard deviation / mean) × 100%

The characteristics of the final dataset presented in Table 2 demonstrate substantial differences in symptom structure between the analyzed medical specialties, with important implications for the design and evaluation of machine learning algorithms in clinical diagnostics. The structure of diagnostic data reflects genuine differences in clinical practice. Pediatrics shows a tendency towards documenting a larger number of subjective symptoms reported by patients or their caregivers. In contrast, internal medicine exhibits a more systematic approach to documenting both the presence and absence of specific symptoms within the differential diagnostic process.

To enable effective modeling of the diagnostic process, a classification structure based on the ICD-10 system was applied, taking into account practical requirements for differentiating disease entities with similar etiology and clinical presentation. The classification strategy involved grouping diagnostic codes representing related disease entities, while codes with unique clinical characteristics were retained as individual categories. This methodology leverages the hierarchical structure of ICD-10, which is inherently complex due to code organization, inclusion and exclusion criteria, and the large number of possible diagnoses [31]. The approach is consistent with prior research on automated ICD-10

coding [31], where multi-label classification necessitated explicit consideration of dependencies between diagnostic codes.

The final classification structure comprised 87 diagnostic classes in total. Thirty-two classes were derived from pediatrics, including 29 code groups and 3 single codes. Fifty-five classes represented internal medicine, comprising 48 code groups and 7 single codes. Each clinical case was represented by a diagnostic feature vector of variable length, with binary values indicating the presence or absence of specific symptoms. This representation provided an effective balance between the diagnostic granularity required for accurate classification and the practical constraints imposed by statistical representativeness of individual categories in machine learning contexts.

After validation procedures, the final dataset exhibited a disease distribution characterized by substantial heterogeneity in diagnostic frequencies, encompassing 369 unique diagnostic codes at the full ICD-10 level. The diagnostic feature space comprised 479 unique clinical parameters, systematically categorized by information source. Sixty-five parameters represented subjective symptoms reported by patients or their legal guardians, 386 represented objective signs identified during standard physical examination, and 28 corresponded to quantitative and qualitative parameters derived from laboratory and imaging diagnostics.

Analysis of data completeness revealed marked interspecialty differences in the availability of clinical information. The mean completeness of diagnostic features was 74.91% in pediatrics and 82.31% in internal medicine, indicating higher documentation quality in the latter. Completeness of primary diagnostic determinations was very high, reaching 99.97% in pediatrics and 100% in internal medicine, confirming the reliability of final diagnoses in the dataset. Demographic data completeness remained high at 94.5% in pediatrics and 97.3% in internal medicine. In contrast, completeness of documentation regarding comorbid conditions differed substantially between specialties, amounting to 22.5% in pediatrics and 41.8% in internal medicine. These differences reflect distinct documentation practices and the specific characteristics of the diagnostic processes inherent to each medical domain.

D. DIAGNOSTIC UNCERTAINTY MODELING MECHANISM

To address the identified challenges associated with the impact of incomplete data on the diagnostic process, a two-stage diagnostic model based on TOP-1 and TOP-3 metrics was developed. The model accounts for uncertainty arising from data incompleteness by generating a ranked list of potential differential diagnoses with associated probabilities. This mechanism was designed in accordance with clinical observations showing that the correct diagnosis is frequently included among the three highest-ranked diagnostic hypotheses, even if it is not the primary output of an automated system. The approach compensates for diagnostic uncertainty by expanding the spectrum of considered diagnoses.

Simultaneously, it preserves a probabilistic hierarchy that reflects the relative credibility of individual diagnostic hypotheses.

1) TWO-STAGE APPROACH (TOP-1 AND TOP-3)

Clinical practice often requires decision-making under conditions of limited information, necessitating consideration of multiple competing diagnostic hypotheses [19]. Empirical studies indicate that physicians routinely apply differential diagnosis, particularly in cases associated with elevated clinical uncertainty. The proposed two-stage methodology reflects these real-world clinical conditions by providing a structured approach to generating diagnostic recommendations. Multidimensional metrics M1, M3 and M5, which assess the system's ability to position the correct diagnosis at different ranks within the differential list, were defined previously in [16]. Their analysis revealed a 65.3% difference in the M1 metric between the most and least effective systems, underscoring the importance of systematic evaluation of expert systems in medical contexts.

In the first stage of TOP-1 methodology, the model identifies the disease entity with the highest posterior probability based on available clinical data. For each diagnostic candidate, a probability value is generated, serving as a quantitative measure of diagnostic uncertainty. Mathematically, for a case represented by a feature vector $x \in R^n$, where n means available diagnostic parameters, the model produces a probability distribution $p \in R^k$ over k possible diagnoses. The probability of diagnosis y_i with given feature x is expressed as $p(y_i | x)$, where the sum of all probabilities is:

$$\sum_{i=1}^k p(y_i | x) = 1 \quad (1)$$

Equation (1) defines the fundamental normalization condition of the probability distribution in the diagnostic space, ensuring that the total probability mass across all possible diagnostic outcomes equals one. Under the TOP-1 approach, the diagnosis with the highest posterior probability is selected in accordance with Equation (2), which represents a standard evaluation criterion in multiclass medical classification tasks [32].

$$\hat{y}^{\text{TOP-1}} = \arg \max_i p(y_i | x) \quad (2)$$

When the predicted probability exceeds the threshold of 0.85, the diagnosis is classified as highly probable. This threshold was established as a conservative criterion representing high model certainty, consistent with confidence thresholds commonly applied in clinical prediction models. The clinical utility of this threshold was verified through sensitivity analysis across values from 0.70 to 0.95, examining the trade-off between prediction accuracy and case coverage.

The second stage of the TOP-3 methodology involves generating a ranked list of the three most probable disease entities along with their relative probabilities, thereby formalizing differential diagnosis as a process based on ranking diseases by posterior probability [32]. The mathematical formulation

of TOP-3, presented in Equation (3), identifies the three diagnoses with the highest probability values:

$$\hat{y}_{\text{TOP-3}} = \{y_{(1)}, y_{(2)}, y_{(3)}\} \quad (3)$$

where $y_{(i)}$ denotes the i -th diagnosis ordered according to decreasing probability, consistent with the probabilistic hierarchy defined in Equation (4):

$$p(y_{(1)} | x) \geq p(y_{(2)} | x) \geq p(y_{(3)} | x) \geq \dots \geq p(y_{(k)} | x) \quad (4)$$

The proposed framework implements comprehensive uncertainty-modeling mechanisms across three complementary levels, reflecting distinct sources of uncertainty encountered in diagnostic practice. Aleatory uncertainty arises from the natural variability of medical data and is taken into account through probabilistic modeling of the distribution of diagnoses in the population, where the system calculates the entropy of the probability distribution as a measure of aleatory uncertainty according to the equation:

$$H(Y | x) = - \sum_{i=1}^k p(y_i | x) \log p(y_i | x) \quad (5)$$

Equation (5) defines Shannon entropy as a measure of aleatory uncertainty in the distribution of diagnostic probabilities, where higher entropy values correspond to greater uncertainty in the diagnostic process. Epistemic uncertainty arises from limited training data or incomplete diagnostic information and is modeled through analysis of prediction variance across different components of ensemble models. Structural uncertainty results from constraints inherent in the adopted mathematical model and is addressed by comparing predictions produced by heterogeneous algorithmic classes, including tree-based, linear, kernel-based and probabilistic models.

The system incorporates a mechanism for adjusting the decision threshold according to the clinical context. This extends standard medical classification approaches by explicitly accounting for the role of probability thresholds in determining the clinical utility of predictive models [33]. In scenarios requiring a high level of diagnostic certainty, particularly for potentially life-threatening disease entities, the system may apply a higher probability threshold for acceptance of the TOP-1 diagnosis, thereby reducing the risk of incorrect decisions under conditions of incomplete data.

The threshold algorithm implements the following decision logic:

- If $p(\hat{y}_{\text{TOP-1}} | x) > \theta_{\text{high risk}}$ and the disease entity belongs to the high-risk category $\rightarrow$ accept TOP-1
- If $p(\hat{y}_{\text{TOP-1}} | x) > \theta_{\text{standard risk}}$ and the case is standard $\rightarrow$ accept TOP-1
- In all other cases $\rightarrow$ proceed to TOP-3 analysis

where $\theta_{\text{high risk}} > \theta_{\text{standard risk}}$ are thresholds calibrated to the specific clinical context.

Special attention in the system design was devoted to validation and prediction quality-control mechanisms, in accordance with the Transparent Reporting of a Multivariable

Prediction Model for Individual Prognosis or Diagnosis + Artificial Intelligence (TRIPOD+AI) guidelines [34]. Verification procedures include: (1) validation of the mathematical consistency of TOP-3 probabilities, (2) assessment of prediction stability across successive executions, and (3) clinical validation of coherence among differential diagnostic outcomes [32].

2) MEDICAL DATA AUGMENTATION STRATEGIES

Rare diseases were underrepresented in the available training data, so we developed a rule-based procedure for generating synthetic training cases. Each generated case was built from ICD-10 diagnostic criteria and symptom co-occurrence patterns drawn from clinical literature, with controlled stochastic variation applied to simulate natural clinical heterogeneity without breaking diagnostic plausibility. All synthetic cases were validated for clinical coherence against established disease presentations.

The augmentation pipeline used three strategies: partial feature shifting (probability 0.3, range $\pm 15\%$), adaptive noise scaled to the clinical importance of each feature, and generation of synthetic feature combinations that preserve inter-feature correlations above 0.7. After augmentation, label distribution deviation stayed below 5% and feature correlations remained above 95% relative to the original data.

Training and test sets were constructed independently. The training data were generated before evaluation data collection, and the two sets share no common patient source. Symptom pattern overlap analysis confirmed this separation: only 0.7% (internal medicine) and 0.3% (pediatrics) of test cases had exact feature combinations matching the training data, which indicates that the models generalized to previously unseen clinical presentations.

Synthetic training cases were validated against established disease presentations through three complementary procedures. First, all rule sets used for case generation were derived from current Polish clinical guidelines and ICD-10 diagnostic criteria, with co-author physicians from the PARROT AI consortium reviewing the rule mappings for clinical plausibility. Second, statistical similarity between synthetic and real-world feature distributions was verified, with feature correlations preserved above 95% and label distribution deviation kept below 5%. Third, symptom-pattern overlap analysis confirmed dataset separation between the synthetic training set and the real-world evaluation set.

3) METHODS FOR ASSESSING PREDICTION UNCERTAINTY

Prediction uncertainty within the expert system was assessed using mechanisms embedded in the applied classification algorithms, combined with a two-stage diagnostic methodology. Each of the 19 analyzed algorithms generates class-specific diagnostic probabilities according to its underlying computational framework. Probabilistic models, such as NB, derive probabilities directly from parametric distributions. Ensemble models estimate probabilities by aggregating

votes from multiple component models. SVM employ Platt scaling to convert decision function outputs into probability estimates. Tree-based models base probability estimates on the proportion of samples within leaf nodes corresponding to individual diagnostic classes. A central component of uncertainty assessment is the TOP-3 approach, in which the system identifies the three most probable diagnoses along with their relative probabilities. The difference between the probabilities of the first and third ranked diagnoses serves as an indicator of diagnostic confidence. Larger differences correspond to higher confidence in the leading diagnosis. The system incorporates a dynamic decision-threshold adjustment mechanism that depends on the clinical context. For high-risk disease entities, higher probability thresholds $\theta_{high\ risk} > \theta_{standard\ risk}$ are applied for acceptance of the TOP-1 diagnosis, thereby increasing diagnostic confidence requirements.

E. IMPLEMENTATION OF CLASSIFICATION ALGORITHMS

1) RATIONALE FOR ALGORITHM SELECTION

The selection of 19 machine learning algorithms was guided by their effectiveness in handling incomplete medical data and by their diverse approaches to modeling relationships between diagnostic features and disease classifications. The selection was based on comparative studies [35] that demonstrated the superior performance of ensemble methods in medical prediction tasks. The results [36] further demonstrated the effectiveness of SVM and RF in medical classification applications. A literature review [37] on the effectiveness of deep learning methods in medical diagnostics highlights the importance of appropriate algorithm selection. The methodological rationale is supported by studies showing that predictive performance can be improved by 8–15% through algorithm choices tailored to specific medical domains and data characteristics.

Reviews of deep learning approaches to medical data imputation indicate that such methods perform best when missing data exceed 30%. In contrast, ensemble algorithms exhibit robustness to data incompleteness due to their capacity to aggregate information from multiple sources and compensate for missing values through informational redundancy.

Based on these assumptions, the final selection encompasses five classes of methods: tree-based, ensemble, linear and probabilistic, distance- and kernel-based, and neural networks. Representatives of each paradigm were selected on the basis of the cited literature [25], [35], [36], [37], ensuring broad methodological coverage and enabling a comparative analysis.

Deep learning architectures with greater representational capacity (transformer-based classifiers, recurrent networks) were not included in the present comparison for two reasons. First, the input representation consists of binary indicators of symptoms and clinical signs without natural sequential or token-level structure that would benefit from attention or recurrence. Second, with 1,655 cases distributed across

87 classes (averaging fewer than 20 per class), parameter-to-sample ratios characteristic of contemporary deep architectures would lead to overfitting risk that exceeds the expected gain. The MLP, which achieved the highest accuracy in pediatrics, represents the neural-network family within a parameter budget appropriate for the dataset size. Scaling to deeper architectures is identified as a direction for future work contingent on substantially larger multicenter datasets.

2) METHODOLOGY FOR SELECTING OPTIMAL HYPERPARAMETERS

The tuning of hyperparameters for medical diagnostic classifiers must balance two conflicting requirements: predictive performance and numerical stability in the presence of incomplete clinical data. The datasets used in this study contain missing diagnostic features and complex relationships between symptoms and diseases; consequently, off-the-shelf, default configurations prove to be insufficient. The optimization procedure must therefore reflect these data characteristics. We employed Bayesian optimization with a tree-structured Parzen estimator (TPE) via the Optuna library. TPE samples the parameter space more efficiently than grid or random search in high-dimensional settings and requires fewer evaluations to converge, which was important given our computational budget. [38].

The optimization pipeline uses MedianPruner for adaptive pruning, which terminates unpromising trials early based on holdout validation performance. Hyperparameter search spaces for all 19 algorithms were defined from published ranges and then adjusted empirically to match the properties of our medical datasets. Representative bounds illustrate the breadth of the search: for Gradient Boosting, `n_estimators` in [50, 300], `learning_rate` in [1e-3, 1.0] on a logarithmic scale, and `max_depth` in [3, 10]; for SVM, the penalty `C` in [1e-3, 1e3] on a logarithmic scale across linear, RBF, polynomial, and sigmoid kernels; and for the Multi-Layer Perceptron, the L2 penalty `alpha` in [1e-5, 1e-1] and the initial learning rate in [1e-4, 1e-1], both on a logarithmic scale. The same TPE sampler (seed 42) was applied to every algorithm, and the Median Pruner was used for the standard-model family. Each algorithm's search space includes parameter consistency checks that prevent unstable configurations from proceeding. Convergence and termination criteria differ per algorithm: standard models ran 100 optimization trials, while SGD was limited to 50 because of its sensitivity to numerical instability. Pruning activated after at least 5 completed trials, used 10 warm-up steps, and evaluated every three steps. Parameter validation runs in parallel with the search itself, so the procedure remains computationally efficient without sacrificing reproducibility of diagnostic outputs.

3) ALGORITHM ADAPTATIONS TO THE SPECIFICITY OF MEDICAL DATA

The application of machine learning algorithms in medical diagnostics requires specialized adaptations that account for

the unique characteristics of clinical data and the complexity of the diagnostic process. In the presented system, a key adaptation involves the implementation of mechanisms for managing data incompleteness. Previous studies [19] have demonstrated the superiority of deep learning approaches for imputing missing values compared with traditional statistical methods. Accordingly, the system adopts a strategic approach aligned with the characteristics of medical data. Binary features representing the presence or absence of symptoms are assigned a default value of “0,” indicating symptom absence. This approach is clinically justified, as undocumented symptoms are treated as absent during diagnostic reasoning. Categorical features are handled through label encoding with imputation of missing values, while mixed-type data are numerically converted with error handling by replacing incomplete values using label encoding. Reference [1] emphasize that data incompleteness can introduce bias into medical predictions.

Histogram-based GB is well suited to medical data because it handles missing values natively, so incomplete diagnostic records do not require separate preprocessing. The implementation uses an optimized binning strategy (max_bins: 128-255), early stopping with validation, and L2 regularization, which together stabilize training on datasets with substantial missing values.

To handle the heterogeneity of medical data, the system uses Dynamic Ensemble Selection. For each test instance, a meta-learning framework evaluates every base classifier on five competence indicators: local accuracy within $k = 7$ nearest neighbors, prediction confidence, neighborhood class distribution entropy, feature space density, and decision margin between the two highest-probability classes. A Random Forest meta-classifier then aggregates these indicators to weight the contributions of five base classifiers (RF, GB, SVM, Gaussian Naive Bayes, and KNN), producing an instance-specific ensemble rather than a fixed voting scheme.

Several standard algorithms were also adapted for this domain. KNN uses a custom feature importance calculation: for each feature, mean distances to nearest neighbors are computed and converted to importance scores through inverse-distance weighting. SVM adaptations depend on kernel type. Linear kernels derive feature importance from coefficient magnitudes, while nonlinear kernels use feature perturbation to assess how individual features affect predictions. AdaBoost iteratively reweights training samples, placing greater emphasis on previously misclassified instances.

The incorporation of interpretability mechanisms represents a critical aspect of adapting algorithms to medical applications, where explainability is essential for clinical acceptance. Integration of the SHAP library enables the explanation of ensemble model predictions by generating SHAP values for each diagnostic feature with support for various model types [39].

Specialty-specific adaptations account for differences between pediatrics and internal medicine by adjusting feature weights according to the characteristics of data within each clinical domain. The system automatically modifies the importance of diagnostic parameters based on their availability and reliability in specific patient populations. Empirical evidence from [40] and [41] confirms diagnostic differences between specialties, demonstrating that pediatric-specific models perform better when applied to pediatric populations. This finding underscores the need for age-adapted machine learning systems.

Pediatrics exhibits greater diagnostic standardization, enabling the use of more precise models with narrower confidence intervals. In contrast, the heterogeneity of cases in internal medicine necessitates more flexible algorithms and broader differential diagnostic mechanisms. Clinical safety is enhanced through the implementation of TOP-3 differential diagnosis, increasing the likelihood of correct diagnosis even when uncertainty exists in the primary prediction. The system further incorporates label format consistency validation mechanisms, ensuring compatibility across different diagnostic representations within the system.

F. METHODOLOGY FOR RESULT EVALUATION

1) METRICS FOR ASSESSING DIAGNOSTIC EFFECTIVENESS

The evaluation methodology for assessing the diagnostic effectiveness of machine learning algorithms in medical applications is a critical component in ensuring the reliability and safety of clinical decision support systems. Given the specific characteristics of medical data and the critical role of diagnostic accuracy, an evaluation framework was developed that extends beyond standard classification metrics commonly used in conventional machine learning applications. The methodological basis followed the updated TRIPOD+AI guidelines, [34] which emphasize the use of clinically relevant measures when evaluating predictive models based on regression or machine learning methods in medical settings.

The first category comprised basic performance metrics, with classification accuracy serving as a fundamental measure of a model's overall effectiveness in correctly identifying disease entities. Accuracy was calculated as the proportion of correct predictions relative to the total number of analyzed cases. In medical applications, particular attention was given to the stability of this metric across different patient subgroups. Consistent with established findings, [42] accuracy alone may be insufficient in medical contexts, as the consequences of diagnostic errors vary depending on the disease entity involved. In the analyzed dataset, accuracy highlighted considerable differences in the effectiveness of various algorithmic approaches across medical specialties.

Diagnostic precision, defined as the ratio of true positive results to the sum of true positive and false positive outcomes, was a key metric for minimizing the risk of incorrect diagnoses. In diagnostic support systems, maintaining high precision is especially important, as false positive diagnoses

may result in unnecessary medical interventions. Such interventions can generate both economic costs and psychological burden for patients.

Diagnostic sensitivity, reflecting the model's ability to identify all true cases of disease, was an equally important evaluation criterion. In medical applications, low sensitivity may lead to missed diagnoses, posing a significant risk to patient safety. The F1-score, representing the harmonic mean of precision and sensitivity, provided a balanced measure of diagnostic performance and was particularly valuable in the context of imbalanced medical datasets. Previous studies [43] indicate that class imbalance is a common challenge in medical applications, as the prevalence of individual disease entities can differ substantially.

The second category of metrics addressed the specificity of the diagnostic process through the TOP-3 performance approach, which reflects real-world clinical practice, where physicians often consider multiple diagnostic hypotheses simultaneously. This methodology was inspired by studies by [16], evaluating symptom checkers using multidimensional metrics to assess a system's ability to position the correct diagnosis at different ranks within a differential diagnosis list.

The third category comprised clinically oriented metrics that account for the nature of medical decision-making. Model calibration represented an additional evaluation dimension, in line with guidelines emphasizing its importance in medical risk prediction models. Calibration refers to the agreement between predicted probabilities and observed diagnosis frequencies and is essential for the interpretability of results in clinical practice. Well-calibrated models enable clinicians to accurately interpret diagnostic probabilities and make informed decisions based on quantitative confidence measures. Model actionability was assessed by evaluating the ability of the system to generate recommendations that support concrete clinical actions. Within diagnostic decision support systems, actionability was measured by analyzing the extent to which clinicians utilized model predictions in their diagnostic processes, with particularly high actionability observed for TOP-3 approaches that provide a ranked list of potential diagnoses along with their relative probabilities.

2) VALIDATION AND ERROR ANALYSIS

The validation strategy employed strict separation between training and test data. Models were trained on synthetic cases generated independently from and prior to the collection of evaluation data. The generation process relied on established medical knowledge from clinical literature and ICD-10 guidelines rather than empirical patterns from the test dataset. Evaluation was performed on a completely independent set of 1,655 authentic clinical cases collected from electronic health records. This synthetic-to-real validation approach addresses practical constraints in medical AI development, where access to large-scale annotated clinical data is limited, while ensuring that reported performance reflects genuine generalization to real-world clinical presentations [44], [45].

Consistent with the observations of [46] regarding temporal evaluation of model performance in medical datasets, the evaluation dataset comprised independently collected clinical cases from routine medical practice. This separation between training and evaluation data prevents information leakage and ensures that reported performance reflects genuine generalization capability.

The validation framework also addressed missing data, which remains a common challenge in clinical datasets. Following established practices [47], sensitivity analysis using imputation techniques was performed. The validation process therefore accounted for how different missing-data handling strategies affected result stability, which was particularly relevant given differences in data completeness across specialties.

Error analysis formed an integral part of the evaluation and included both quantitative and qualitative assessment of classification errors. In line with the methodology by [48] for reducing errors in clinical decision support systems, a systematic categorization of errors was conducted according to error type and potential clinical impact. This included identification of algorithm-specific error patterns and assessment of whether errors exhibited systematic or random characteristics. The confusion matrix was the primary analytical tool, enabling detailed examination of classification errors for each algorithm and providing insight into specific difficulties in discriminating between pairs of disease entities. Qualitative error assessment additionally included analysis of borderline cases, identifying cases with the greatest disagreement between algorithms and the lowest classification confidence. In accordance with the recommendations of [49], these cases underwent additional clinical review to identify factors contributing to diagnostic difficulty.

The adequacy of the sample size was verified in accordance with the guidelines set out in [50] regarding the validation of machine learning methods in clinical settings. The available sample size was sufficient to detect differences in efficacy at a specified significance level while maintaining adequate statistical power. To assess whether differences in accuracy between algorithms were statistically significant, McNemar's paired tests with exact p-values were used. Confidence intervals (95%) for TOP-1 and TOP-3 accuracy were calculated using bootstrap resampling with 1,000 iterations.

IV. RESULTS

Diagnostic accuracy was assessed on the basis of 616 clinical cases in internal medicine and 1,039 cases in pediatrics. The performance of the algorithms varied significantly depending on the model. To ensure that the classifiers had not simply memorized examples from the training data, we measured the degree of overlap in feature patterns between the training and test sets: only 0.7% of test cases in internal medicine and 0.3% of cases in pediatrics exhibited identical combinations of symptoms to the training data. The models therefore demonstrated the ability to generalize to previously unknown clinical presentations. Detailed results are presented in Table 3.

TABLE 3. Comparative analysis of the diagnostic effectiveness of machine learning algorithms under TOP-1 and TOP-3 evaluation approaches.

Algo rithm	TOP -1 Accu racy (%)	95% CI (TOP- 1)	TOP- 3 Perfor mance (%)	95% CI (TO P-3)	Preci sion (%)	F1- Sc ore (%)	Classifi cation error (%)
Internal Medicine (n=616)							
GB	84.74	(81.82–87.50)	90.91	(88.64–93.19)	86.47	84.34	15.26
RF	83.12	(80.19–86.04)	90.58	(88.31–92.86)	85.43	82.91	16.88
SVM	82.95	(80.03–85.88)	91.07	(88.80–93.34)	84.13	82.78	17.05
MLP	82.63	(79.55–85.39)	91.72	(89.45–93.83)	84.01	82.57	17.37
KNN	82.31	(79.38–85.23)	87.82	(85.23–90.26)	84.33	82.10	17.69
Extra Tree s	81.82	(78.90–85.06)	90.42	(87.99–92.69)	84.36	81.21	18.18
Dyna mic Ense mble	81.66	(78.57–84.90)	91.40	(89.12–93.67)	83.51	81.32	18.34
Hist GB	81.49	(78.25–84.74)	90.42	(88.15–92.86)	83.28	80.94	18.51
LR	80.84	(77.92–83.77)	88.31	(85.71–90.75)	81.96	80.75	19.16
Pediatrics (n=1039)							
ML P	87.20	(85.27–89.22)	92.78	(91.24–94.32)	87.46	87.15	12.80
Extra Tree s	85.85	(83.73–87.97)	93.94	(92.59–95.38)	87.34	85.73	14.15
KNN	84.22	(81.91–86.53)	93.17	(91.72–94.80)	85.39	84.21	15.78
Hist GB	83.54	(81.14–85.85)	93.65	(92.01–95.09)	84.36	83.34	16.46
SVM	82.77	(80.27–85.08)	93.84	(92.40–95.28)	83.96	82.46	17.23
Dyna mic Ense mble	82.19	(79.88–84.41)	94.71	(93.36–96.15)	82.98	82.19	17.81

TABLE 3. (Continued.) Comparative analysis of the diagnostic effectiveness of machine learning algorithms under TOP-1 and TOP-3 evaluation approaches.

RF	80.85	(78.34–83.35)	94.42	(93.07–95.77)	86.36	80.80	19.15
Extra Tree	78.34	(75.84–80.85)	83.06	(80.85–85.27)	78.82	78.36	21.66
GB	77.29	(74.59–79.79)	93.46	(92.01–94.90)	80.33	76.88	22.71

Notes: Bold values indicate the best results within a given metric category.

Bootstrap 95% confidence intervals were computed using 1,000 resamples to assess the precision of accuracy estimates. Pairwise McNemar tests evaluated the statistical significance of performance differences between algorithms. In internal medicine, differences among the four top-performing algorithms (Gradient Boosting, Random Forest, SVM, and MLP) were not statistically significant ($p > 0.05$), with overlapping confidence intervals (79.55–87.50%). Gradient Boosting significantly outperformed all algorithms ranked fifth or lower ($p < 0.05$). The highest TOP-1 performance was achieved by GB, which also exhibited the lowest classification error. This finding supports the advantage of sequential ensemble methods for complex diagnostic problems characteristic of internal medicine. The iterative learning process, whereby the algorithm learns from predecessor errors, is particularly valuable when modeling heterogeneous clinical data typical of this specialty.

In pediatrics, MLP and Extra Trees showed no significant difference ($p > 0.05$), while MLP significantly outperformed KNN and all lower-ranked algorithms ($p < 0.05$). The Multi-Layer Perceptron (MLP) achieved the highest TOP-1 accuracy and the lowest classification error, indicating the advantage of neural networks in pediatric diagnostics. This result represents a notable improvement over traditional ensemble methods and confirms the ability of MLP to model complex diagnostic patterns in pediatric populations. Neural network architectures facilitate identification of nonlinear relationships between symptoms, which is especially relevant in pediatrics, where clinical presentation may differ depending on the patient’s developmental stage.

A detailed analysis of GB error patterns (Figure 3) indicates characteristic clusters of misclassifications within related disease entities. The algorithm performs best for the following codes: J06, J06.8, J06.9, and J00 (acute upper respiratory tract infection; 77 cases), followed by M50, M50.1, M50.9, M50.8, M54.5, M54.4, M54.3, M54.1, M54, M48, M47.9, M47, M53.1 (spinal disorders; 36 cases), and M25.5, M79.6, M79.1, M20.1, M20.5, M89.9, M21.4 (limb and joint pain symptoms and foot deformities; 35 correct classifications). Classification errors were rare and occurred primarily

as isolated instances (1–2 cases), without clear clusters of confusion across related entities. This dispersed error structure, in which most off-diagonal entries were 0, supports the algorithm's capacity for precise differentiation between disease entities. The limited number of cross-system errors is particularly important in internal medicine, where multimorbidity is common.

High performance was also observed for RF and SVM, supporting the utility of diverse algorithmic approaches in internal medicine diagnostics. RF, as a classical ensemble method, reduces overfitting risk by aggregating multiple decision trees, which may be particularly beneficial when medical data are incomplete. SVM, although not an ensemble approach, achieved the third-best result due to its ability to model complex decision boundaries in a high-dimensional feature space.

Analysis of the MLP confusion matrix (Figure 2) reveals characteristic classification patterns in pediatrics. Diagonal dominance is particularly evident for the most common pediatric diagnoses, including J06.9, J06.8, J06, J06.0 (306 correct), J00 (99 correct), J20.9, J20, J22, J40 (59 correct), and J02.0, J02, J02.9 (50 correct). These results confirm the ability of MLP to effectively recognize the dominant respiratory conditions in the pediatric population. Error analysis nonetheless indicates systematic confusion patterns within respiratory diagnoses. The most prominent misclassifications occur between J00 (acute nasopharyngitis) and the J06.9, J06.8, J06, J06.0 group (acute upper respiratory infections), with 17 misclassified cases. Similarly, J18 and J18.9 (pneumonia) are misclassified as J06.9, J06.8, J06, or J06.0 in five cases, which may reflect difficulties in distinguishing between milder and more severe infections based on initial clinical presentation. A notable positive finding is the minimal number of cross-system errors, with virtually no substantial confusion between J codes (respiratory diseases) and H codes (ear diseases), B codes (infectious diseases), or R codes (general symptoms). This supports the ability of MLP to distinguish effectively between major organ systems. High performance was also observed for Extra Trees and KNN, indicating the value of diverse algorithmic approaches in pediatric diagnostics. Extra Trees, through the introduction of additional randomization during decision tree construction, demonstrates increased resistance to overfitting while maintaining high predictive performance. KNN, grounded in a case-similarity paradigm, confirms its suitability for pediatrics, where greater case homogeneity favors methods based on local similarity.

Analysis of TOP-3 performance is particularly important in evaluating diagnostic decision support systems, as it reflects real-world clinical practice based on differential diagnosis. In this context, a substantial improvement in performance was observed compared with the single-step approach, with a mean TOP-3 effectiveness of 77.43% across all algorithms. MLP achieved the highest TOP-3 effectiveness at 91.72%. These findings indicate considerable potential to enhance

diagnostic safety through the incorporation of differential diagnostic mechanisms in clinical decision support systems.

TOP-3 performance analysis also revealed exceptionally high effectiveness in pediatrics, with a maximum value of 94.71% achieved by the Dynamic Ensemble approach. These results demonstrate strong algorithmic potential for identifying alternative diagnostic hypotheses in pediatric settings. The mean TOP-3 effectiveness across all algorithms in pediatrics was 81.99%, representing a 4.57 percentage point improvement compared with internal medicine. These observations suggest greater diagnostic standardization and more predictable symptom presentation in pediatric populations, potentially attributable to lower comorbidity rates and more homogeneous clinical manifestations in this age group.

Analysis of diagnostic confidence across individual algorithms (Figure 4) revealed varying capabilities in recognizing different diagnostic categories. High confidence was consistently observed for the following codes: H61.2, H93.1, H60.5, H60.9, H60, H66.0, H66.9, H65.1, H65.2, H65.0, H65, H90.3, H90, H91 (ear disorders) and I10, I11 (hypertension), with most algorithms, including GB and Extra Trees, achieving confidence scores above 0.90. This consistently high confidence suggests well-defined and characteristic symptomatology for these conditions in the internal medicine population. In contrast, probabilistic algorithms (AdaBoost, Bernoulli NB, and Multinomial NB) consistently exhibited the lowest confidence values, typically ranging from 0.3 to 0.6 across most diagnostic categories. This pattern confirms the limited applicability of classical probabilistic methods and simple boosting approaches when dealing with complex, high-dimensional internal medicine data characterized by incomplete features.

Analysis of the diagnostic confidence heatmap for pediatrics (Figure 5) revealed similarly characteristic patterns. High confidence was observed for the following codes: H66.4, H66.0, H66, H66.9, H65, H65.0, H60, H60.9 (otitis media), B07, B08, B08.5, B08.4, B01, B01.9, B00 (viral skin diseases) and J02.0, J02, J02.9 (upper respiratory tract infections), with most algorithms achieving confidence scores above 0.80. This consistently high confidence for common pediatric conditions further confirms greater symptom standardization in this population. Again, probabilistic algorithms (AdaBoost and Bernoulli NB) showed systematically lower confidence values across most groups. In contrast, MLP, Extra Trees, and KNN demonstrated a balanced confidence profile, with high values across the majority of disease entities. Notably, low confidence values were observed for J10, J10.1, J11, J15.9, J98.8, J98.9 codes (influenza and pneumonia codes), even among the best-performing algorithms. This likely reflects the inherent difficulty of differentiating more severe infections from milder respiratory conditions based on the available clinical features.

Diagnostic precision analysis showed consistently high values for ensemble algorithms, with GB (86.47%) and RF (85.43%) achieving the highest scores. Precision is critically



In pediatrics, precision values were likewise high, with the strongest results observed for MLP (87.46%) and Extra Trees (87.34%). High precision is especially important in

Finally, analysis of classification error distributions showed that algorithms with the highest TOP-1 effectiveness also exhibited the lowest error rates, confirming consistency across evaluation metrics. GB (15.26% error) and RF (16.88% error) demonstrated the highest diagnostic reliability, a critical consideration for the practical deployment of clinical decision support systems, where minimizing

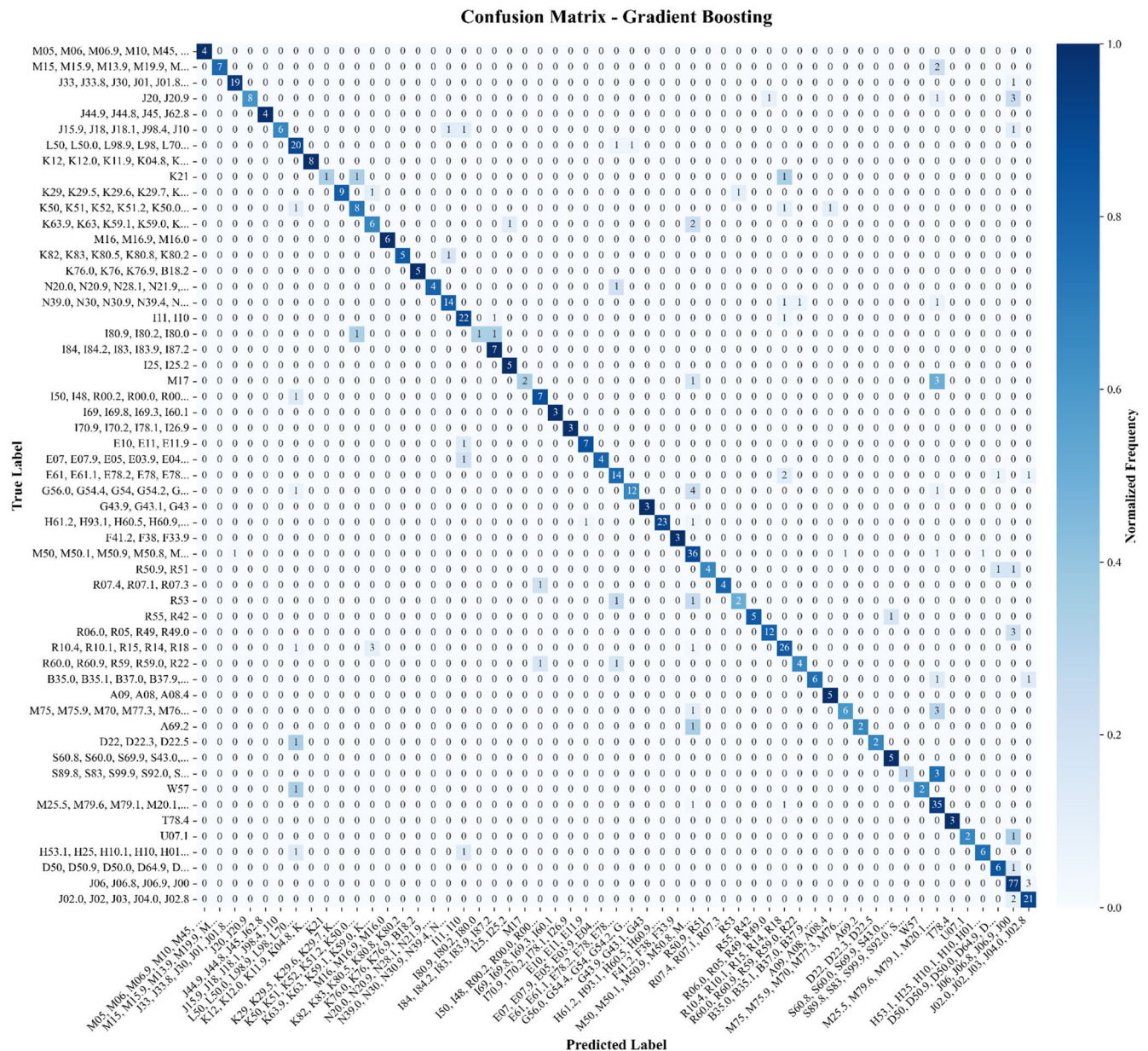


FIGURE 3. Confusion matrix of the GB algorithm in internal medicine. Diagonal values represent correct classifications; off-diagonal values represent misclassifications between ICD-10 codes.

misdiagnosis risk is a primary patient safety objective. This error rate corresponds to an incorrect first suggestion in approximately one in six cases. However, upon application of TOP-3, the risk of missing the correct diagnosis falls below 9%, representing a clinically significant improvement in diagnostic safety.

A consolidated overview of the basic evaluation metrics is presented in Figure 6, confirming the dominance of ensemble methods in the heterogeneous environment of internal medicine.

Analysis of classification errors in pediatrics demonstrated lower error rates than those observed in internal medicine,

with the minimum error of 12.80% achieved by MLP. This improvement is clinically meaningful, as each percentage point reduction in classification error directly enhances patient safety and strengthens clinician confidence in diagnostic decision support systems. At the same time, the range of classification errors in pediatrics (12.80%–22.71%) was broader than in internal medicine, reflecting greater variability in diagnostic processes within this specialty. The application of the TOP-3 mechanism reduces the risk of missing the correct diagnosis to below 6%, underscoring the clinical safety advantage of differential diagnosis ranking in this specialty. A consolidated summary of the basic

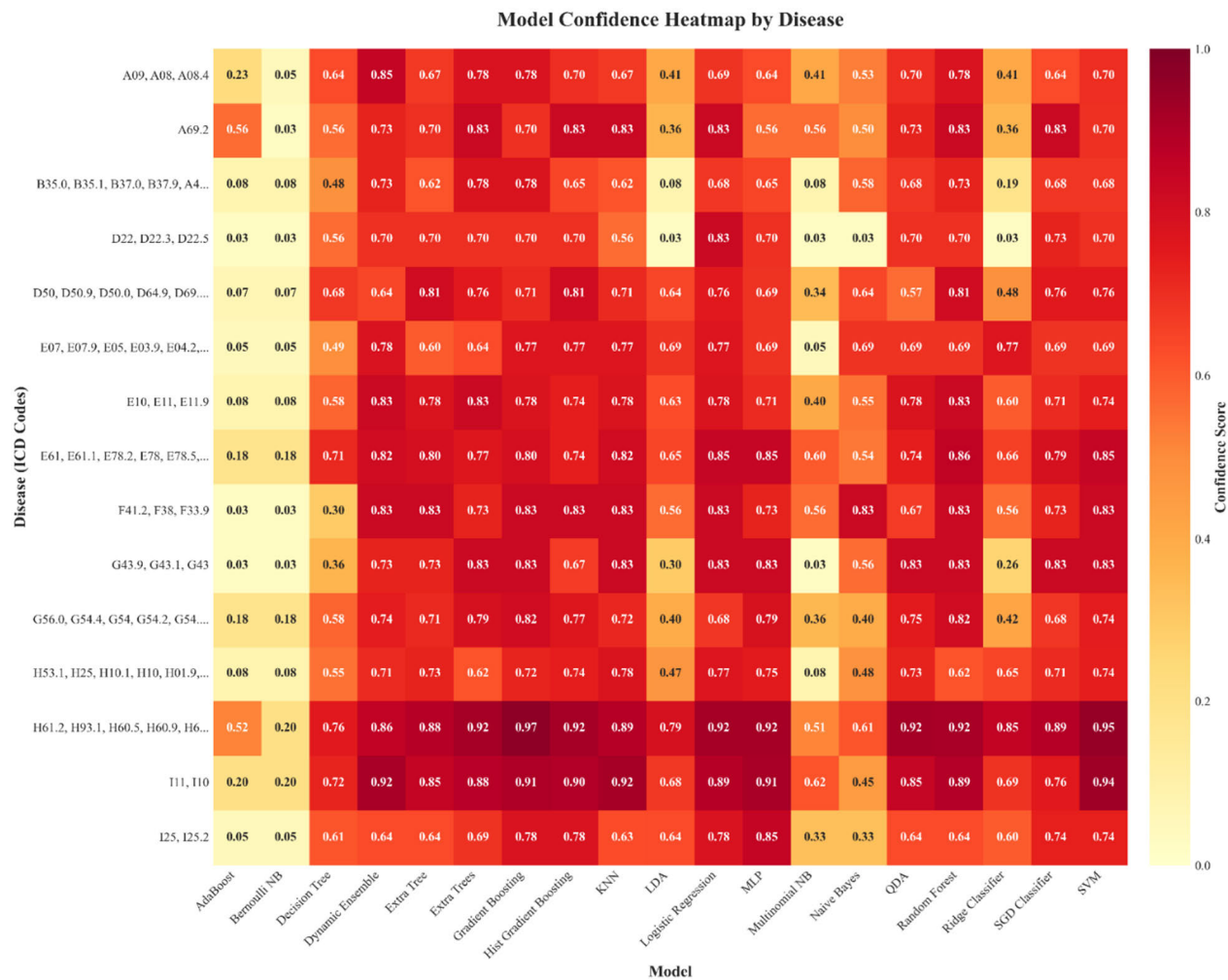


FIGURE 4. Heatmap illustrating diagnostic confidence of algorithms across ICD-10 categories in internal medicine.

performance metrics is presented in Figure 7, confirming the advantage of neural networks and algorithms with increased randomization in the more standardized pediatric setting.

Analysis of TOP-3 performance revealed a substantial clinical advantage of the evaluated models across both specialties. Even when the top-ranked prediction was incorrect, the probability of identifying the correct diagnosis among the three highest-ranked results exceeded 94% for the best-performing models in pediatrics and 91% in internal medicine. This marked improvement over the single-step approach is of fundamental importance for the clinical safety of decision support systems and may contribute to reducing diagnostic errors in situations of incomplete clinical information. Analysis of the prediction agreement matrix (Figure 8) highlights fundamental differences in the decision-making structures of algorithms across specialties. The most pronounced discrepancy was observed for AdaBoost, which in internal medicine exhibited extremely low agreement with all other algorithms (0–0.06), whereas in pediatrics agreement

increased to 0.31–0.70. This finding suggests that, in internal medicine, AdaBoost produces diagnostic decisions that differ substantially from those of other algorithms, partially explaining its very low effectiveness (5.52%). In internal medicine, a clear polarization of agreement patterns was observed: algorithms either formed tightly clustered groups with high mutual agreement (for example, GB, Extra Trees, RF) or exhibited very low agreement, particularly in the case of probabilistic methods (NB algorithms). In contrast, the agreement structure in pediatrics was more homogeneous, with even the weakest algorithms (AdaBoost, Bernoulli NB) maintaining moderate agreement with others. In pediatrics, MLP showed high agreement with ensemble methods (0.86–0.88), whereas in internal medicine this agreement decreased to 0.64–0.80. This suggests that, in pediatrics, different algorithmic paradigms tend to converge on similar diagnostic solutions. More diffuse decision boundaries in pediatric data indicate that greater data standardization leads to convergent modeling strategies. Conversely, the heterogeneity of internal medicine data necessitates greater

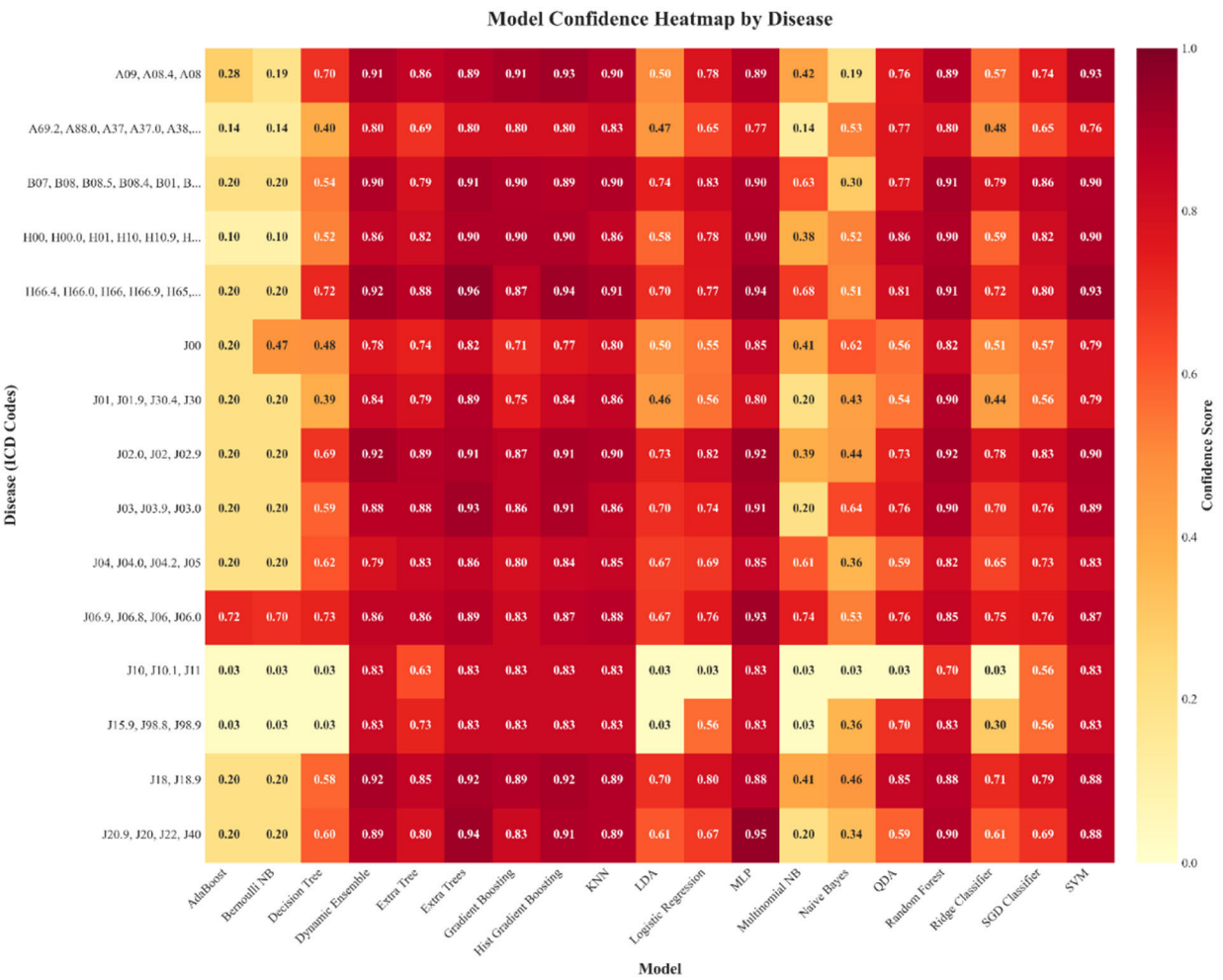


FIGURE 5. Heatmap illustrating diagnostic confidence of algorithms across ICD-10 categories in pediatrics.

algorithmic specialization, resulting in increased divergence in diagnostic decision-making.

Sensitivity analysis of the 0.85 confidence threshold confirmed its clinical utility across both specialties. Predictions exceeding this threshold demonstrated substantially higher accuracy for the best-performing ensemble algorithms compared to overall model performance. The threshold corresponds approximately to the 75th percentile of prediction confidence for correctly classified cases, ensuring that high-confidence predictions represent the upper quartile of model certainty while maintaining clinically meaningful case coverage.

The implementation of the TOP-3 mechanism in routine clinical practice may function as an early warning system in cases of high diagnostic uncertainty, where low confidence in the primary prediction signals the need to consider alternative diagnostic hypotheses. Moreover, probability rankings within the TOP-3 framework can provide valuable insight into the relative likelihood of competing diagnoses, supporting clinical decision-making in complex diagnostic scenarios.

An interpretability analysis was conducted for the most effective models in both specialties: GB in internal medicine (n = 616) and MLP in pediatrics (n = 1039), using SHAP [39] and LIME. For both models, KernelExplainer was applied to a background sample compressed using the k-means method. SHAP values were calculated for the full test set in pediatrics and a balanced sample of correct and incorrect predictions in internal medicine. The LIME analysis was performed for all incorrect predictions and a sample of correct predictions in both specialties. Global SHAP values revealed different feature hierarchies between specialties. In pediatrics, respiratory symptoms (cough, rhinitis, nasal congestion) dominated, which corresponds to the epidemiological profile of this population. The MLP also assigned high importance to normal physical examination findings (normal blood pressure, normal vesicular breath sounds), indicating that the network learned to use the absence of pathology as a diagnostic signal, which is interesting and offers opportunities for further research in the future. Normal pulmonary auscultation argues against pneumonia and directs the prediction toward milder

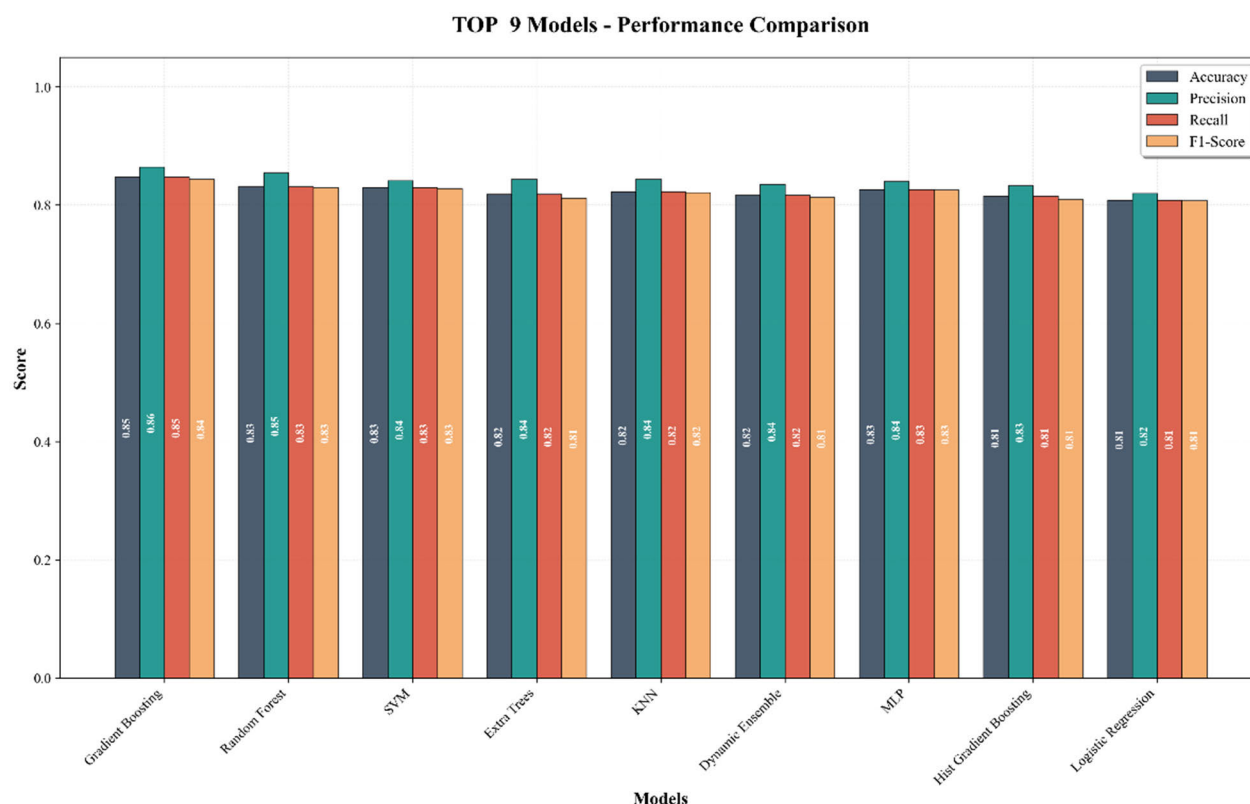


FIGURE 6. Comparison of basic performance metrics of the nine best-performing algorithms in internal medicine.

infections. In internal medicine, GB identified a more diverse set of features (spinal pain, referral to orthopedist, rhinitis, fever, hypertension), reflecting greater heterogeneity in cases.

Heatmaps of normalized SHAP feature importance for the TOP-25 features, stratified by ICD-10 category (Figures 9 and 10), allow verification of whether the learned patterns correspond to established medical knowledge. In pediatrics (Figure 9), a clear block structure is evident: ear pain activates almost exclusively for otitis media (H66), soft abdomen predominates in gastroenterological conditions (K59), and COVID-19 forms an isolated block at U07.1. In internal medicine (Figure 10), the structure is more diffuse: spinal pain and orthopedic referral concentrate on M-codes (musculoskeletal disorders), hypertension activates selectively for I10/I11, and cardiology referral activates for cardiac conditions (I25, I50). The model utilizes ordered diagnostic tests (referral ECG, referral X-ray) as indirect indicators of clinical suspicion.

Feature importance heatmaps reveal clinically consistent patterns of diagnostic assignments across both specialties. In pediatrics (MLP, $n = 1,039$), the highest normalized SHAP values concentrate within specific ICD-10 categories, confirming the alignment of features with their corresponding ICD-10 categories. Normal blood pressure and normal vesicular breath sounds predominate in respiratory infection categories (J06, J10/J11), rhinitis and erythematous throat exhibit the highest importance for acute nasopharyngitis

(J00), and cough-related symptoms (dry cough, productive cough) are most important for acute laryngitis and tracheitis (J04). Soft abdomen attains the highest importance in the functional gastrointestinal disorder category (K59/K30), while COVID-19 is the exclusively dominant predictor of category U07.1. Notably, fever exhibits diffuse importance across multiple diagnostic categories, reflecting its nonspecific clinical character. In internal medicine (GB, $n = 616$), the heatmap reveals more pronounced feature specialization. Spinal pain and orthopedic referral predominate in musculoskeletal disease categories (M25/M79/M20), hypertension and ECG referral concentrate in cardiovascular disease categories (I11/I10), and sinusitis and leg pain exhibit the highest importance for category J33 and M-group conditions, respectively. Normal urinalysis disperses across multiple categories, serving as an exclusionary feature analogously to fever in pediatrics. Specialist referrals (to dermatologist, cardiologist, surgeon) form distinct diagnostic blocks associated with corresponding ICD-10 groups, confirming the model's capacity to leverage referral information in the diagnostic process. SHAP-LIME cross-validation demonstrated agreement of 43.1% for GB and 31.9% for MLP (proportion of shared features in the TOP-5). Both approaches identified clinically meaningful features, differing in their respective rankings. The observed values are consistent with the literature, where agreement of 30–50% for nonlinear models reflects the methodological complementarity of the two approaches.

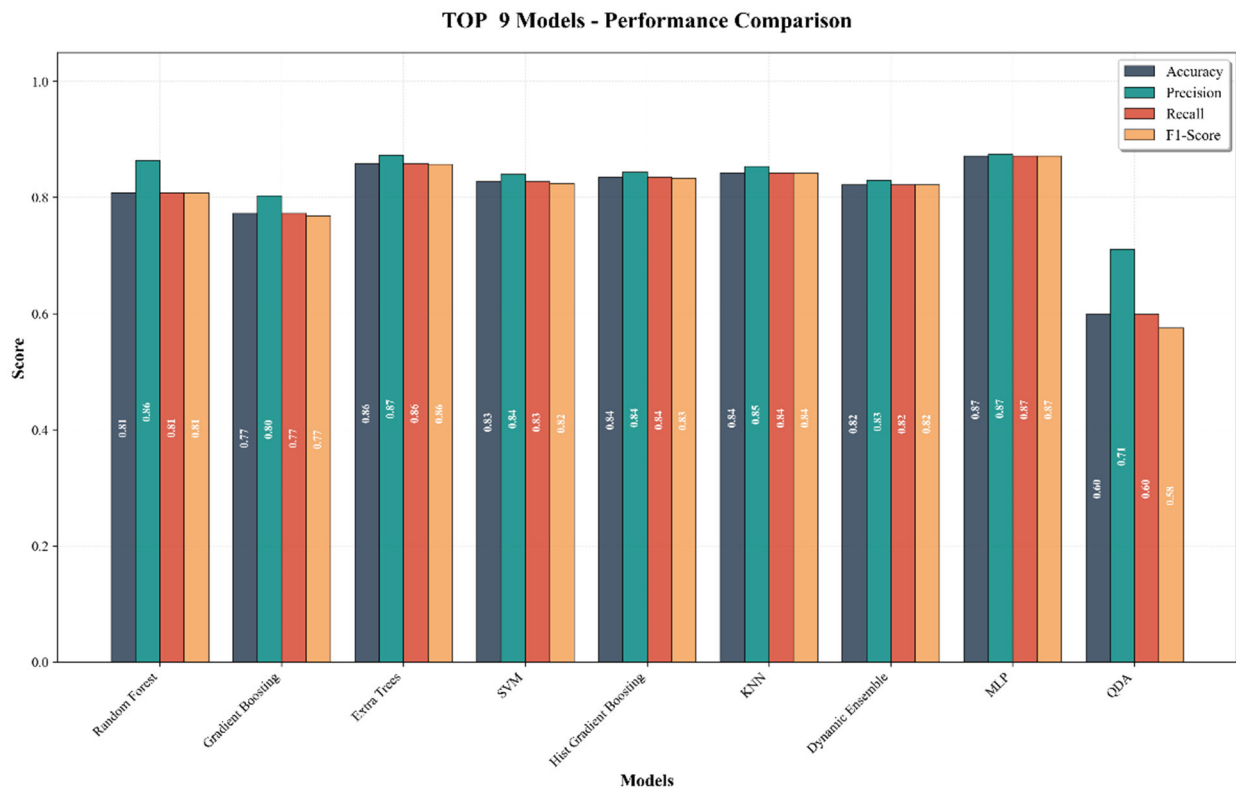


FIGURE 7. Comparison of basic performance metrics of the nine best-performing algorithms in pediatrics.

Clinical validation of the SHAP feature-importance heatmaps was performed by co-author physicians from the PARROT AI consortium. For each specialty, the top 25 features per ICD-10 category were qualitatively assessed for consistency with established diagnostic patterns. Agreement was rated as fully consistent for 22 of 32 categories in pediatrics and 38 of 55 categories in internal medicine, with the remaining categories rated as partially consistent, driven by feature granularity rather than discordant clinical reasoning. No category showed contradiction with established medical knowledge.

To verify the generalization capability of the models, external validation was conducted on a set of 120 clinical cases (60 from internal medicine and 60 from pediatrics) collected from a medical facility located in a different region of Poland than the original test set. These cases were entirely independent of both the training data and the original evaluation set of 1,655 cases. The results are presented in Table 4.

TABLE 4. Results of external validation on 120 independently collected clinical cases from a healthcare facility in a different region of Poland.

Algorithm	TOP-1 Accuracy (%)	TOP-3 Performance (%)	Precision (%)	F1-Score (%)
Internal Medicine (n=60)				
GB	73.33	86.67	76.02	73.43
Pediatrics (n=60)				
MLP	78.33	90.00	85.42	79.83

TOP-1 accuracy was lower than in the original evaluation (73.33% vs. 84.74% in internal medicine; 78.33% vs. 87.20% in pediatrics), which is an expected effect of external validation on data from a different facility and a different patient population. TOP-3 accuracy remained at a high level (86.67% in internal medicine, 90% in pediatrics), confirming the clinical utility of the differential diagnosis mechanism under conditions differing from those of training. The tendency toward higher accuracy in pediatrics was preserved.

To assess whether aggregate performance metrics are influenced by class prevalence, the 87 diagnostic categories were stratified by case-count tier. Twelve categories contained at least 30 test cases (high-prevalence tier), 28 contained between 10 and 29 cases (medium-prevalence tier), and 47 contained fewer than 10 cases (low-prevalence tier, with a minimum of 3 cases per class consistent with the inclusion criterion). Pooled across both specialties, micro-average TOP-1 accuracy decreased across tiers (88.8%, 83.7%, and 81.9%, respectively), indicating that aggregate accuracy partially reflects the prevalence structure of the dataset. Performance estimates for the lowest-prevalence classes should accordingly be interpreted as exploratory. The full per-class distribution and its tier-level aggregation are provided in the Supplementary Material (Tables S1 and S2).

To address potential label leakage from referral-type features, an ablation analysis was performed for the best-performing model in each specialty. Each model was retrained on a feature subset excluding all referral indicators

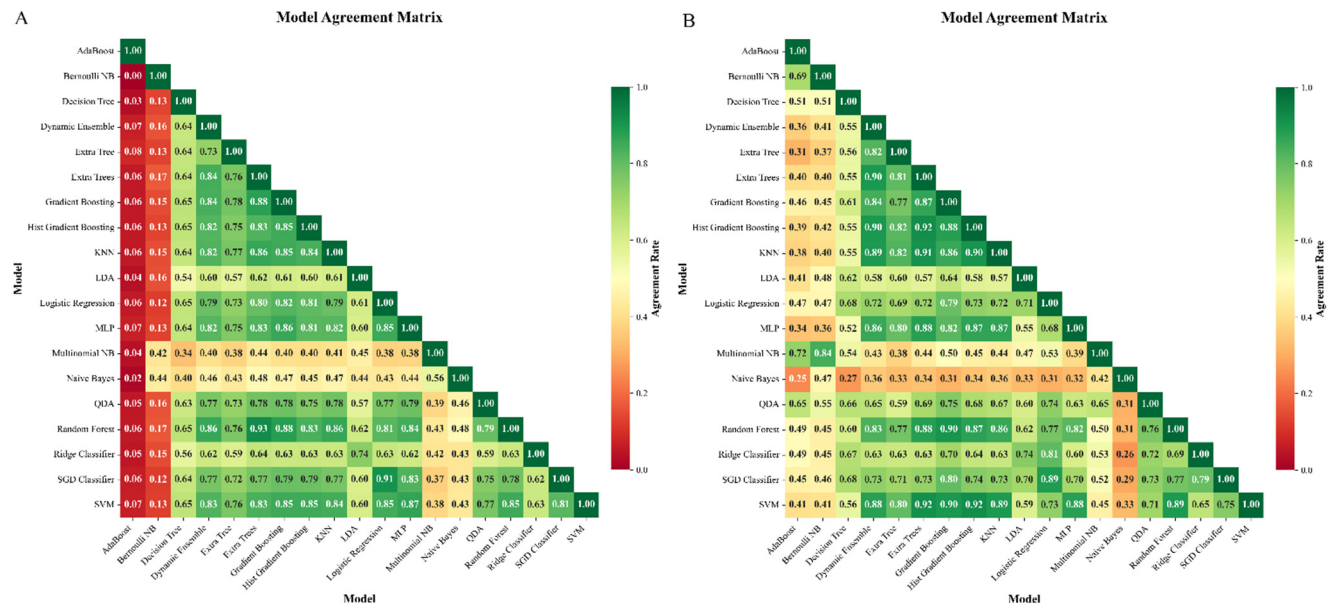


FIGURE 8. Prediction agreement matrix between algorithms in internal medicine (A) and pediatrics (B).

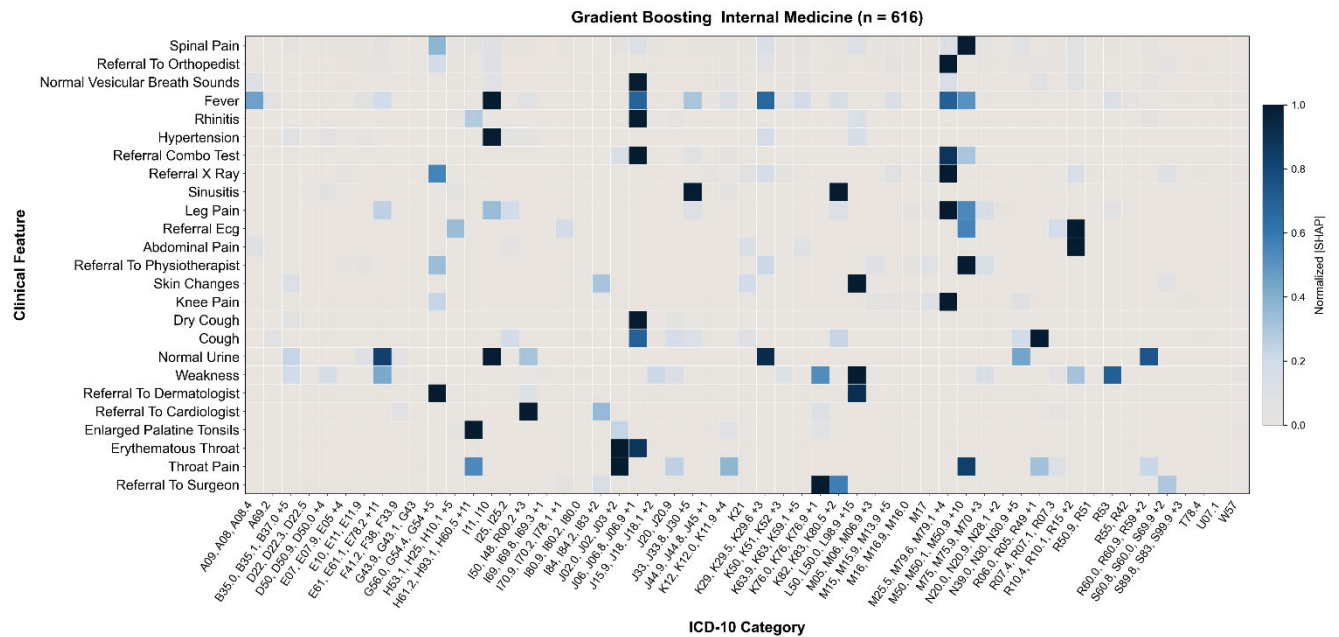


FIGURE 9. Normalized SHAP feature importance stratified by ICD-10 category, GB, internal medicine (n = 616).

(specialist referrals, ordered laboratory and imaging examinations), using the same Optuna-optimized pipeline and the same internal evaluation set as the primary analysis. In pediatrics, the MLP achieved TOP-1 87.20% and TOP-3 92.78%, identical to the full-feature configuration ($\Delta = 0.0$ p.p.). In internal medicine, the GB model achieved TOP-1 80.68% and TOP-3 90.10%, compared with 84.74% and 90.91% in the full-feature configuration ($\Delta = -4.06$ p.p. on TOP-1 and -0.81 p.p. on TOP-3). The absence of a substantial performance collapse upon removal of referral features indicates that the models do not rely on referrals as a proxy

for the final diagnosis. The small, measurable contribution in internal medicine is consistent with the clinical role of referrals (e.g., cardiology, orthopedic, ECG) as indirect indicators of clinical suspicion, whereas in pediatrics, dominated by symptom-based respiratory diagnoses, referral features contribute no measurable predictive value. In both specialties, predictive performance is preserved without referral features, supporting the conclusion that predictions are driven by symptoms rather than by ordered diagnostic actions.

The intended deployment of the diagnostic classification system within the PARROT AI platform is as a

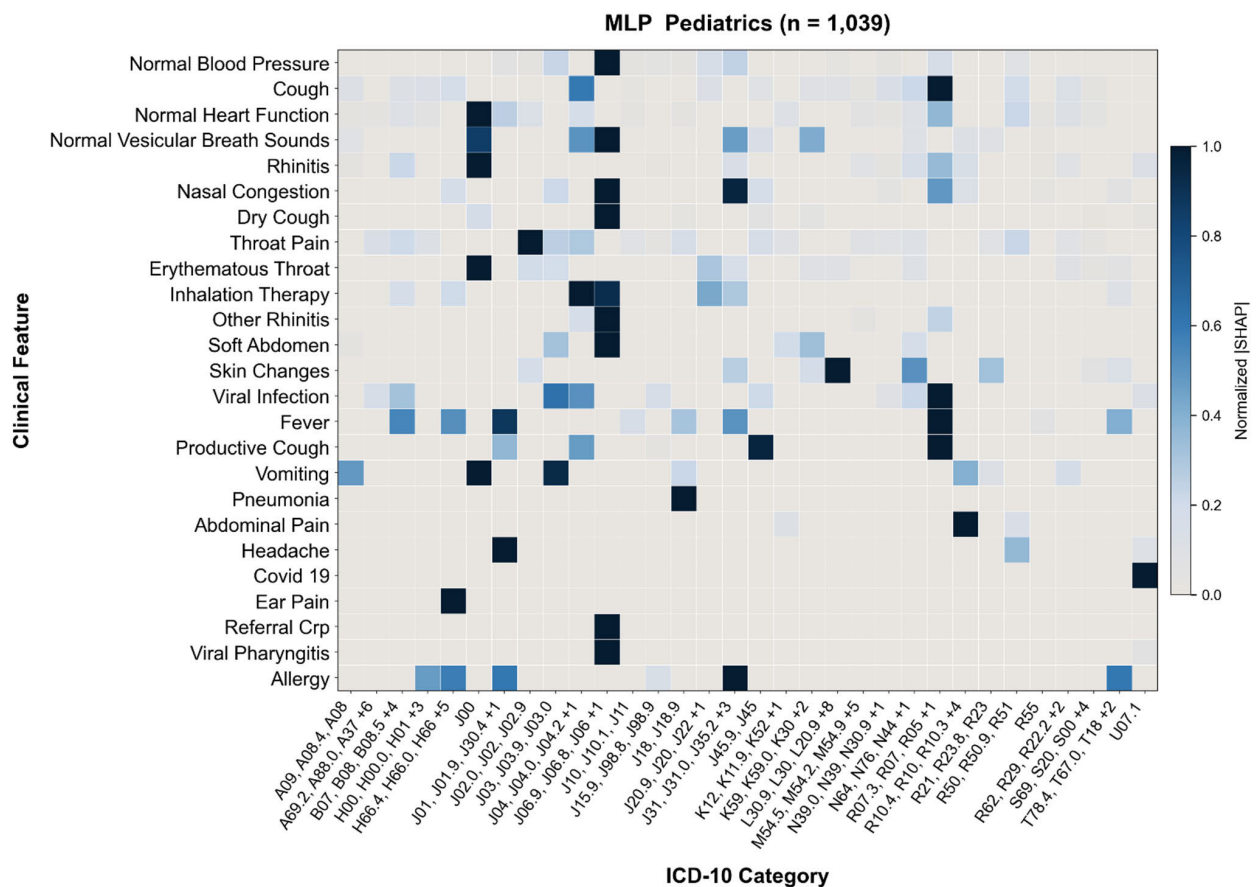


FIGURE 10. Normalized SHAP feature importance stratified by ICD-10 category, MLP (n = 1039).

post-consultation decision-support module, invoked after the physician has completed history taking and physical examination; for real-time decision support during the consultation itself, a referral-free model variant is recommended, and the ablation analysis quantifies the expected performance trade-off.

V. DISCUSSION

An analysis of 1,655 clinical cases from pediatrics (n = 1,039) and internal medicine (n = 616) demonstrates the applicability of machine learning algorithms in medical diagnostics characterized by incomplete clinical data. The results are consistent with the hypothesis that ensemble algorithms can partially compensate for diagnostic uncertainty through differential diagnosis mechanisms, as reflected in the observed performance differences between medical specialties. Direct quantification of the relationship between data completeness levels and diagnostic accuracy remains an area for future investigations.

A key finding is the specialty-dependent variability in algorithm performance. In internal medicine, GB achieved the best results, with a classification error of 15.26%, whereas in pediatrics the MLP was superior, with a classification error of 12.80%. This divergence indicates fundamental

differences in the structure and characteristics of diagnostic data between the two specialties, in line with theoretical assumptions regarding domain-specific diagnostic processes. The statistical analysis revealed distinct patterns of algorithm differentiation between specialties. Internal medicine exhibited a cluster of four statistically equivalent top-performing algorithms (McNemar $p > 0.05$), suggesting that multiple modeling approaches achieve comparable effectiveness in this heterogeneous clinical domain. In contrast, pediatrics showed clearer algorithm stratification, with MLP demonstrating statistically significant superiority over most competitors ($p < 0.05$).

The two-stage diagnostic mechanism (TOP-1 and TOP-3) reflects real-world clinical practice based on differential diagnosis. For the best-performing models, the probability of identifying the correct diagnosis among the three highest-ranked predictions was 94.71% (95% CI: 93.36–96.15%) in pediatrics for Dynamic Ensemble, and 91.72% (95% CI: 89.45–93.83%) in internal medicine for MLP. This improvement is fundamental to clinical safety, as it reduces the risk of missing the correct diagnosis in settings characterized by incomplete clinical information. Furthermore, the analysis of TOP-3 performance demonstrated a substantial increase in diagnostic effectiveness, representing

an important contribution to the design of clinical decision support systems.

A comparative analysis consistently showed higher diagnostic accuracy in pediatrics across all evaluated metrics, with a mean TOP-3 performance advantage of 4.57 percentage points (81.99% vs 77.43%). This superiority may be attributed to greater diagnostic standardization in pediatrics. In contrast, internal medicine exhibited greater case heterogeneity, resulting in a broader range of diagnostic challenges. These findings have practical significance and highlight the need to tailor strategies for implementing diagnostic support systems to the specific characteristics of each medical specialty.

The results have direct implications for the design of clinical decision support systems. The statistical equivalence of the top-performing algorithms in internal medicine indicates that algorithm selection in this specialty should take into account interpretability and inference efficiency. The clear superiority of MLP in pediatrics justifies investment in neural network infrastructure in this domain. The higher coefficient of variation in internal medicine (51.8% vs. 48.9% in pediatrics) quantitatively confirms the need for more flexible algorithms in this specialty. These observations argue in favor of a modular system architecture with specialty-specific models, rather than a single universal classifier.

The multi-layer medical data augmentation methodology applied in this study proved essential for maintaining clinical credibility while increasing training data diversity. The augmentation pipeline, incorporating partial feature shifting (probability 0.3, range $\pm 15\%$), controlled adaptive noise, and synthetic combination generation, preserved high data fidelity, with label distribution deviation below 5% and feature correlations exceeding 95%. The high diagnostic accuracy achieved on real clinical data (exceeding 84%) provides indirect validation of the synthetic training data quality, confirming that augmented cases preserved clinically meaningful patterns.

The findings have direct relevance for the development of clinical decision support systems in real-world medical practice. The probability difference between the first and third diagnosis in the TOP-3 ranking serves as a quantitative confidence indicator, enabling automatic identification of high-uncertainty cases that require clinical verification. This mechanism is particularly valuable in facilities with limited specialist availability, where structured support for differential diagnosis may improve diagnostic outcomes. A similar mechanism was employed in [12], where cases with low prediction confidence were automatically flagged for expert review. This confirms the feasibility of implementing multi-stage diagnostic mechanisms in clinical decision support systems.

The interpretability analysis confirms that the models base their predictions on clinically relevant features. For correct predictions, SHAP explanations point to symptoms constituting the established diagnostic basis for a given diagnosis. For incorrect predictions, SHAP transparently reveals the

underlying causes of errors, enabling the clinician to critically evaluate the system's suggestions in the context of an individual case. SHAP analysis, available across all classes in both pediatrics and internal medicine, constitutes a form of qualitative validation complementary to quantitative performance metrics.

External validation on 120 cases from an independent facility in a different region of Poland demonstrated a decline in TOP-1 accuracy compared to the original evaluation. This decline is typical of external validation and stems from differences in patient population, documentation style, and the diagnostic profile of the facility. TOP-3 accuracy remained at 86.67–90%, meaning that the correct diagnosis appears among the three proposed candidates in nearly 9 out of 10 cases. This confirms the practical utility of the system as a tool supporting differential diagnosis even under conditions differing from those of training. Full multicenter and international validation remains a direction for future research.

It is important to acknowledge that the TOP-1 accuracy values of 84.74% (internal medicine) and 87.20% (pediatrics) reported on the internal evaluation set reflect synthetic-to-real transfer rather than full clinical generalization. The training data were generated from ICD-10 criteria with rule-based augmentation, and the evaluation set was labeled using the same ICD-10 framework. This methodological alignment may inflate measured performance relative to fully external clinical conditions. Accordingly, the external validation results, TOP-1 73.33% (internal medicine) and 78.33% (pediatrics) on 120 cases from an independent facility, should be regarded as the primary benchmark of clinically meaningful performance. The TOP-3 accuracy of 86.67–90% on external validation is consistent with the system's intended role as a differential-diagnosis support tool rather than an autonomous diagnostic device.

The methodological strategy of this study, training on rule-based synthetic data validated against clinical guidelines and evaluating on real-world cases, was chosen as a pragmatic response to the constraints of obtaining adequately sized labeled datasets for 87 diagnostic classes in Polish primary care. We do not advocate this approach as universally preferable to training on real clinical data; for high-prevalence specialties with established large-scale labeled corpora, training on real data remains the gold standard. The synthetic-training pathway is best understood as a feasibility approach for early-stage development of differential-diagnosis support systems, suitable for hypothesis generation and architectural validation but not for substantiating claims of clinical utility. Regardless of the training-data approach, no claim of clinical utility can be substantiated without an adequately powered prospective study using real clinical data from multiple facilities and varied clinical settings, evaluated against established physician-performance benchmarks on identical cases.

The SHAP–LIME top-feature agreement of 31.9% (MLP, pediatrics) and 43.1% (GB, internal medicine) requires careful methodological interpretation. SHAP and LIME formalize different notions of feature attribution: SHAP computes

Shapley values reflecting each feature's marginal contribution to the model's output across all coalitions, while LIME approximates the model locally with a linear surrogate around a single prediction. Disagreement between these methods is a well-documented phenomenon in the explainable AI literature [51], [52], [53], with rank-correlation values for nonlinear models typically in the range of 0.19–0.50 across diverse datasets. Our observed 30–50% top-K feature overlap is consistent with this body of evidence and reflects methodological complementarity rather than internal inconsistency of the diagnostic model. From a clinical standpoint, the recommended interpretation is the following: features identified by both methods (the intersection) should be regarded as robust evidence supporting the prediction; features identified by only one method should be treated as auxiliary signals warranting verification against the patient's clinical context. We do not recommend that clinicians weigh SHAP and LIME outputs independently. The system instead presents a consolidated view in which the SHAP global ranking serves as the primary explanation, with LIME used selectively for case-level audit when prediction confidence is low or contradicts the clinician's initial impression.

Several limitations should be considered when interpreting these results. The analysis was based on data from a single healthcare system, potentially limiting generalizability to other populations and organizational contexts. The case selection methodology may have excluded rare diseases, thereby narrowing the diagnostic scope of the analyzed models. The requirement of a minimum of three cases per diagnostic code may have eliminated clinically relevant but infrequent conditions; performance estimates for the retained low-prevalence classes ($n < 10$) should accordingly be interpreted as exploratory and require confirmation in larger multicenter datasets. Moreover, the study focused primarily on diagnostic accuracy and did not fully address factors such as time to diagnosis, diagnostic costs, or effects on patient outcomes. The clinical data additionally contain systematic biases. The difference in documentation completeness between pediatrics (74.91%) and internal medicine (82.31%) is systemic in nature, and the inclusion criteria may have favored more thoroughly documented cases. The ICD-10 labels reflect the decisions of individual physicians, and the impact of these biases on the predictive fairness of the system has not been investigated. The synthetic data used for model training may not fully capture atypical clinical presentations, generating only textbook disease manifestations. The study does not address challenges associated with real-world clinical deployment, such as physician acceptance of the system, integration with electronic health records, regulatory requirements, and concept drift. External validation conducted on 120 cases from a single additional facility confirms generalization. However, the sample size is limited. Multicenter validation at a larger scale, as well as on data from foreign healthcare systems, is necessary for a comprehensive assessment of the clinical applicability of the system.

The results obtained are consistent with existing research on machine learning applications in medical diagnostics, with the effectiveness of ensemble algorithms corresponding to observations [35]. Similar findings were reported in a comparative evaluation of nine ML algorithms for cervical cancer classification [10]. Tree-based ensemble methods (RF, LightGBM, CatBoost) consistently outperformed the other evaluated approaches. It should be noted, however, that this study concerned a single disease on imputed data. The present work operates on real-world incomplete clinical data encompassing 87 diagnostic categories across two specialties, constituting a substantially more complex classification task. These findings collectively demonstrate the superiority of ensemble methods in disease prediction and confirmatory findings [36] demonstrating the particular effectiveness of SVM and RF in medical classification tasks. Observed differences between specialties align with previous findings [40] indicating higher effectiveness of pediatrics-specific models.

The developed system achieved TOP-1 accuracies of 87.20% in pediatrics and 84.74% in internal medicine within a domain-specific diagnostic space of 87 ICD-10 categories. This design reflects the intended deployment context, where the system augments specialist physicians within defined clinical workflows rather than serving as a general-purpose triage tool. The diagnostic module operates within the broader PARROT AI system [7], which integrates solutions for contact center solutions, automatic speech recognition, and semantic analysis of medical interviews, providing comprehensive support for clinical processes. The text classification component in this processing chain [14] addresses the challenge of extracting structured clinical information from noisy dialogue data, serving as the input layer for the diagnostic reasoning mechanism presented in this work.

A systematic review and meta-analysis of 83 studies [15] reported an overall diagnostic accuracy of 52.1% for generative AI models. These models did not differ significantly from non-specialist physicians (difference 0.6%, $p = 0.93$), but performed significantly worse than experts (difference 15.8%, $p = 0.007$). In a subsequent study [54], 20 LLMs were systematically evaluated across 264,800 diagnostic assessments covering 1,655 real-world clinical cases. Standalone LLMs achieved only 16–20% TOP-1 accuracy, and integration with RAG yielded a minimal improvement (+2–5 percentage points). In contrast, machine learning-based expert systems improved performance by approximately a factor of 2.5 [54]. Interestingly, the full integration of LLM, RAG, and expert system components paradoxically reduced accuracy compared to the expert system alone, indicating that naive combination of heterogeneous diagnostic sources may degrade rather than improve performance. These observations are further supported by [55], which directly compared a dedicated expert system with the GPT-4 and Gemini 1.5 models based on 36 unpublished clinical cases, demonstrating that the expert system made the correct diagnosis more often than both LLMs. This confirms

the enduring clinical significance of machine learning-based decision support systems, especially in situations with incomplete diagnostic data. These results prompted the authors of this study to focus on optimizing machine learning-based classification independently of LLM model architectures. While direct comparison is limited due to differences in task scope (constrained 87-class classification vs. open-domain diagnosis), the developed system achieved 84–87% accuracy across both specialties, suggesting competitive performance within the structured classification setting. A study of a diagnostic system in pediatrics [9] reported a mean F1-score of 0.885, exceeding that of junior physicians (0.841 and 0.839) while remaining below that of senior physicians (0.907–0.923). The developed system achieved F1-scores of 0.872 in pediatrics and 0.847 in internal medicine. While direct comparison with physician-level performance is not warranted given the differences in evaluation protocol, case mix, and clinical setting, these values position the system as a competitive automated classifier within its 87-class structured task. A formal head-to-head comparison with practicing physicians on identical cases is identified in Future Work as a necessary next step.

The performance of systems operating on complete datasets provides an important reference point. A symptom-based diagnostic system study [8] showed that, on synthetic datasets containing a full set of symptom features, selected models achieved accuracy and F1-scores of up to 99%. The developed system, maintaining accuracy above 84% despite data incompleteness, demonstrates the capacity for effective operation in real clinical practice. This contrast underscores a key implementation challenge, as complete clinical data are rarely available.

From a theoretical standpoint, the findings support the effectiveness of different machine learning strategies depending on medical data characteristics. Informational redundancy among ensemble model components appeared particularly beneficial in internal medicine (GB, RF), whereas the detection of complex non-linear patterns was crucial in pediatrics (MLP, Extra Trees). These observations align with theoretical assumptions regarding matching algorithms to data-specific properties. The inter-specialty differences observed suggest that diagnostic process characteristics vary in ways that extend beyond the simple complexity of clinical cases. Higher diagnostic standardization in pediatrics may reflect not only reduced case heterogeneity, but also age-specific symptom development and distinct patterns of clinical presentation.

VI. CONCLUSION AND FUTURE WORK

This study presents the application of machine learning algorithms in medical diagnostics characterized by incomplete clinical data. The analysis of 1,655 clinical cases from two medical specialties confirmed the main research hypothesis by demonstrating that appropriately selected algorithms can effectively compensate for diagnostic uncertainty arising from data incompleteness through the implementation of

differential diagnostic mechanisms. GB achieved the highest effectiveness in internal medicine (classification error 15.26%, McNemar $p > 0.05$ for RF, SVM, MLP). MLP was dominant in pediatrics (classification error 12.80%, McNemar $p < 0.05$ for most competitive algorithms). RF and Extra Trees algorithms demonstrated high precision values (85.43% and 87.34%, respectively), confirming their effectiveness in interpreting incomplete medical data. A key finding was the significantly higher diagnostic effectiveness observed in pediatrics compared with internal medicine, with mean TOP-3 performance values of 81.99% and 77.43%, respectively – difference 4.57 p.p. The proposed two-stage diagnostic mechanism (TOP-1 and TOP-3) represents a methodological contribution to modeling diagnostic uncertainty, with TOP-3 performance analysis revealing the clinical advantage of the evaluated models. Even when the first prediction was incorrect, the probability of identifying the correct diagnosis among the three highest-ranked results exceeded 94.71% (Dynamic Ensemble) in pediatrics and 91.72% (MLP) in internal medicine, substantially enhancing the clinical safety of the decision support system.

The study contributes methodologically through the development of a multidimensional algorithm evaluation framework incorporating TOP-3 analysis and clinical safety assessment strategies. The applied rule-based synthetic data generation methodology enabled model training under data scarcity constraints while achieving high diagnostic accuracy on authentic clinical cases, demonstrating the feasibility of knowledge-informed training data generation for diagnostic support systems. The empirical contribution includes a comparison of 19 classification algorithms using real clinical cases with varying levels of informational completeness, providing valuable insight into the practical effectiveness of individual methods. External validation on 120 clinical cases from an independent medical facility in a different region of Poland is treated as the primary benchmark of the system's real-world performance, with TOP-1 accuracy of 73.33% in internal medicine and 78.33% in pediatrics and TOP-3 accuracy of 86.67% and 90.00% respectively. The internal-set results reported above are accordingly regarded as an upper-bound reference under synthetic-to-real transfer rather than as evidence of full clinical generalization. The identification of different optimal algorithms for each specialty confirms the necessity of tailoring methods to data-specific characteristics.

The findings have direct relevance for the design of clinical decision support systems, with the TOP-3 mechanism offering potential implementation as an early warning system in cases of high diagnostic uncertainty. Probability ranking provides clinicians with a hierarchical list of potential diagnoses, supporting informed clinical decision-making. The identified inter-specialty differences highlight the need to adapt algorithms to the specific characteristics of each medical field. Pediatrics, characterized by higher diagnostic standardization, may benefit from more precise models, whereas the heterogeneity of cases in internal medicine

necessitates algorithms with greater flexibility and broader differential diagnostic mechanisms. From a computational perspective, inference time was not systematically evaluated in this study. However, all implemented algorithms including gradient boosting, random forests, support vector machines, and multi-layer perceptrons operate within millisecond-scale response times on standard clinical workstation hardware. Given that typical diagnostic consultations span several minutes, computational latency is negligible for practical clinical deployment and does not constitute a barrier to real-time decision support. The integration of the developed diagnostic module into the PARROT AI system [7], combined with text classification based on an ensemble method for noisy clinical dialogues [14], enables the creation of a complete process from transcribing the patient interview to issuing a diagnostic recommendation.

In future, this methodology should be tested in specialties other than internal medicine and pediatrics, particularly in fields with a different diagnostic logic, such as psychiatry, radiology and emergency medicine. Rare diseases present a distinct challenge: training data is by definition limited, so effective modeling will likely require the use of data augmentation techniques, specialized imputation for conditions with very low prevalence.

The application of few-shot learning or transfer learning may be critical for effective modeling of such rare disorders. Preliminary external validation confirmed the effectiveness of the system on data from a different facility and patient population. The next step is to extend validation to a larger number of centers, additional medical specialties, and data from foreign healthcare systems, which will allow confirmation of the reproducibility of results and an analysis of the impact of systematic differences in data completeness on the diagnostic performance of the system. A key direction for future work involves conducting prospective clinical studies to assess the real-world impact of diagnostic decision support systems, considering not only diagnostic accuracy but also time to diagnosis, healthcare costs, and the satisfaction of patients and medical personnel. Integration with electronic medical record systems will require the development of application programming interfaces for real-time clinical data processing, along with continuous learning mechanisms enabling models to adapt to evolving clinical patterns. It is also necessary to improve the methodology for generating synthetic data in order to better represent atypical clinical presentations. A complementary direction is the integration of severity-weighted evaluation metrics that reflect the differing clinical consequences of individual error types, alongside cost-utility analysis appropriate to the deployment setting.

Federated learning may facilitate model development on large-scale clinical datasets while preserving data privacy. The integration of multimodal data, including textual, imaging, laboratory, and physiological signal data, necessitates the development of unified structures capable of efficiently processing heterogeneous medical information under varying levels of data completeness. Future studies should also

address the ethical implications of artificial intelligence in medical diagnostics, including algorithmic bias, system transparency, and accountability in cases of diagnostic error.

The results confirm the hypothesis that ensemble-based approaches can mitigate diagnostic uncertainty, and the two-stage methodology has demonstrated consistent effectiveness for both internal medicine and pediatrics data, despite their differing characteristics. The differences we observed between specialties suggest that clinical decision support systems should not treat all medical fields in the same way, particularly when training data is limited or the clinical picture is ambiguous.

VII. DATA AND CODE AVAILABILITY

The clinical data and source code forming the basis of this study are not publicly available. Both the dataset and the implementation are subject to the consortium agreement of the PARROT AI project, funded by the National Centre for Research and Development under the INFOSTRATEG IV strategic research and development program (grant agreement INFOSTRATEG-IV/0012/2022). The program is a research and commercialization initiative, involving contractual obligations to implement the resulting technologies as commercial products, and the intellectual property management plan approved by the funding body excludes the public release of the dataset and the implementation code. Methodological reproducibility is supported by detailed descriptions of all algorithms, hyperparameter search spaces, training and evaluation procedures, and synthetic data generation rules presented in Section III. External validation on 120 independent clinical cases from another institution provides empirical confirmation of generalization that does not depend on access to the training code or data.

VIII. TERMINOLOGY

To ensure consistent use of terminology, the following terms are applied with distinct meanings throughout the manuscript. Missing values denote absent entries at the level of individual features in the diagnostic feature vector, together with their algorithmic handling. Missing data denotes the statistical mechanism of absence (e.g., MAR, MNAR) and the patterns by which entries are absent. Data incompleteness (and the adjective incomplete clinical data) denotes the property of the dataset as a whole, namely that real-world clinical records are inherently incomplete; the taxonomy of its types (systemic, cognitive, procedural, temporal, semantic, disproportionate) is defined in Section III. Diagnostic uncertainty denotes uncertainty at the level of model prediction, quantified through the posterior probability distribution and the TOP-3 mechanism, and is further decomposed into aleatory, epistemic, and structural uncertainty.

LIST OF ABBREVIATIONS

Abbreviation	Full name
AdaBoost	Adaptive Boosting.
AI	Artificial Intelligence.
API	Application Programming Interface.

BioBERT	Bidirectional Encoder Representations from Transformers for Biomedical Text Mining.
CDSS	Clinical Decision Support System(s).
CKD	Chronic Kidney Disease.
CLI	Command Line Interface.
EHR	Electronic Health Record.
Extra Tree	Extremely Randomized Trees.
GB	Gradient Boosting.
GUI	Graphical User Interface.
Hist GB	Histogram-based Gradient Boosting.
ICD-10	International Classification of Diseases, 10th Revision.
ICD-10-CM	International Classification of Diseases, 10th Revision, Clinical Modification.
JSON	JavaScript Object Notation.
KNN	k-Nearest Neighbors.
LDA	Linear Discriminant Analysis.
LR	Logistic Regression.
M1	Metric@1 (TOP-1 accuracy).
M3	Metric@3 (TOP-3 accuracy).
M5	Metric@5 (TOP-5 accuracy).
MAR	Missing At Random.
ML	Machine Learning.
MLP	Multi-Layer Perceptron.
MNAR	Missing Not At Random.
NB	Naive Bayes.
NDCG	Normalized Discounted Cumulative Gain.
NLP	Natural Language Processing.
PDF	Portable Document Format.
REST	Representational State Transfer.
RF	Random Forest.
ROC-AUC	Receiver Operating Characteristic – Area Under the Curve.
SGD	Stochastic Gradient Descent.
SHAP	SHapley Additive exPlanations.
SVC	Support Vector Classifier.
SVM	Support Vector Machine.
TOP-1	Top-1 accuracy (true diagnosis ranked 1st).
TOP-3	Top-3 accuracy (true diagnosis within top 3).
TOP-5	Top-5 accuracy (true diagnosis within top 3).
TPE	Tree-structured Parzen Estimator.
TRIPOD+AI	Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis + Artificial Intelligence.
Web API	Web Application Programming Interface.
XLSX	Excel Open XML Spreadsheet file format.

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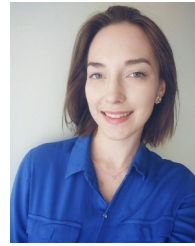
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